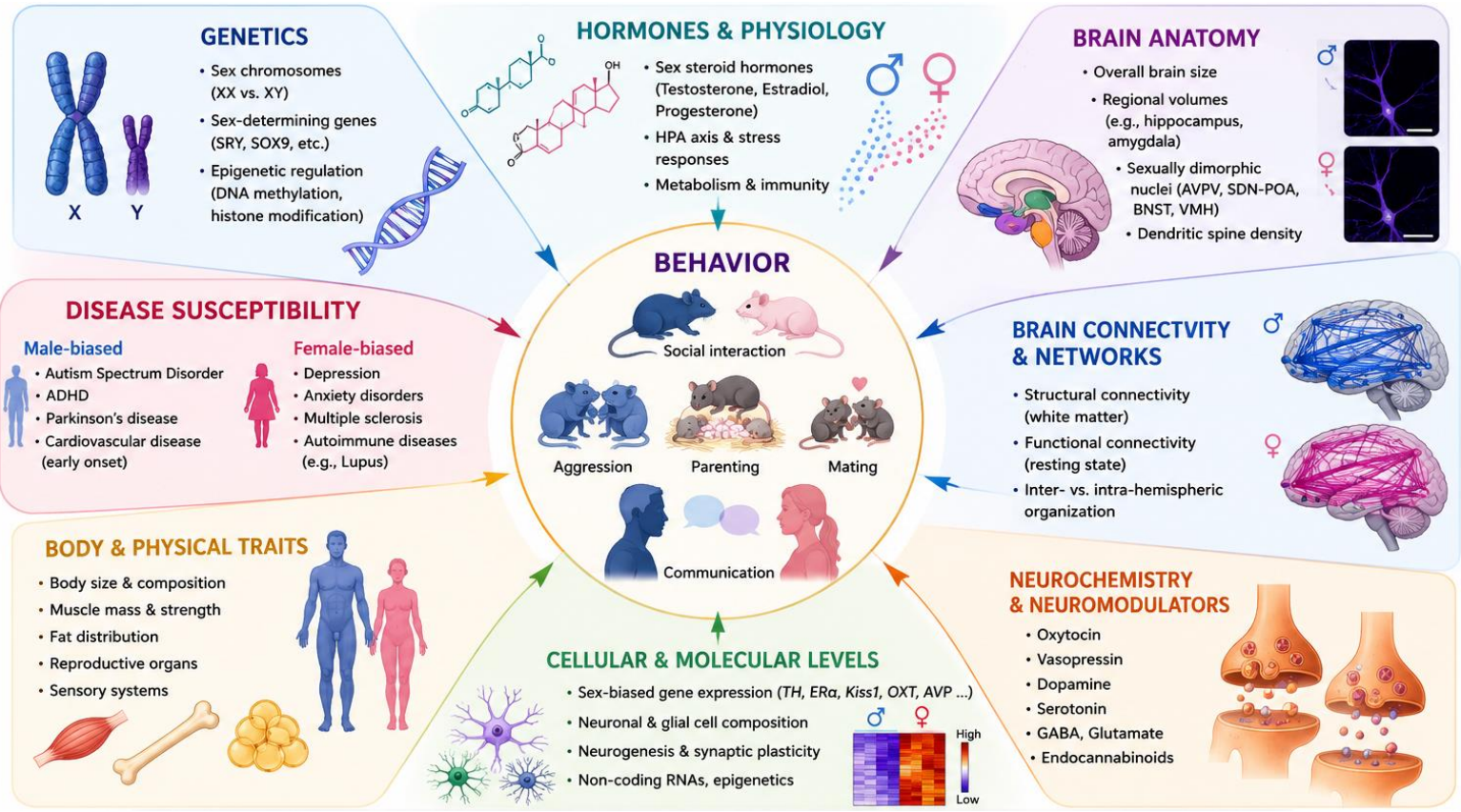
Behavioral Neuroscience

Sex differences in brain and social behavior: From genes to circuits



June, 2026

Why study sex difference (sex as a variable)



Sex differences emerge from the dynamic interplay of genes, hormones, brain, body and environment ultimately shaping behavior and health

Social behavior in mammalian species



- **Essential for survival and reproduction of all species**
- **Complex behavior controlled by multiple genes and environmental factors**

Sex differences in (innate) social and reproductive behaviors



Territoriality (aggressive behavior)



Parental care



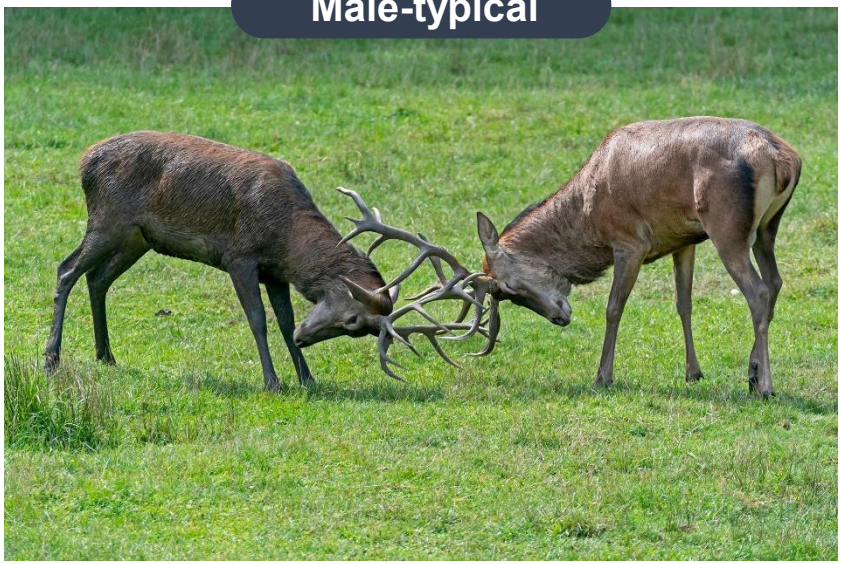
Courtship behavior



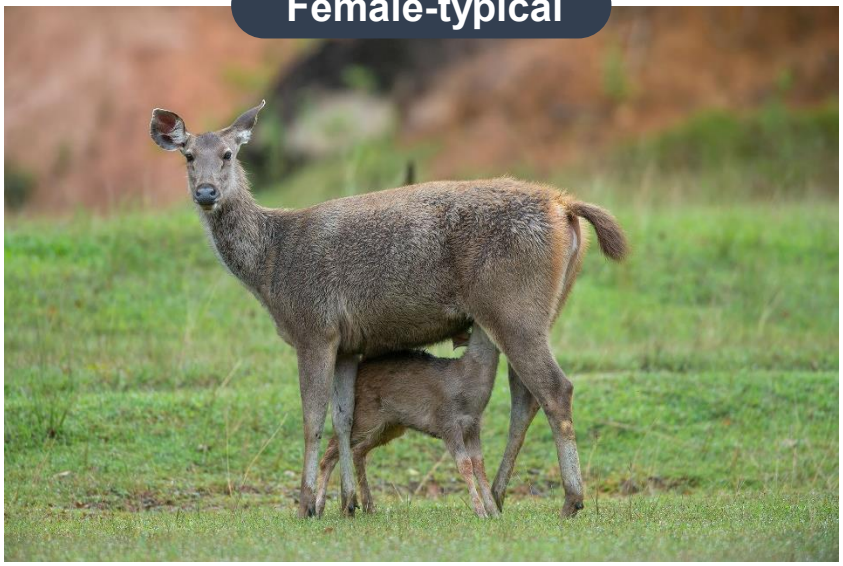
Sexual behavior

Sex-typical social and reproductive behavior

Male-typical

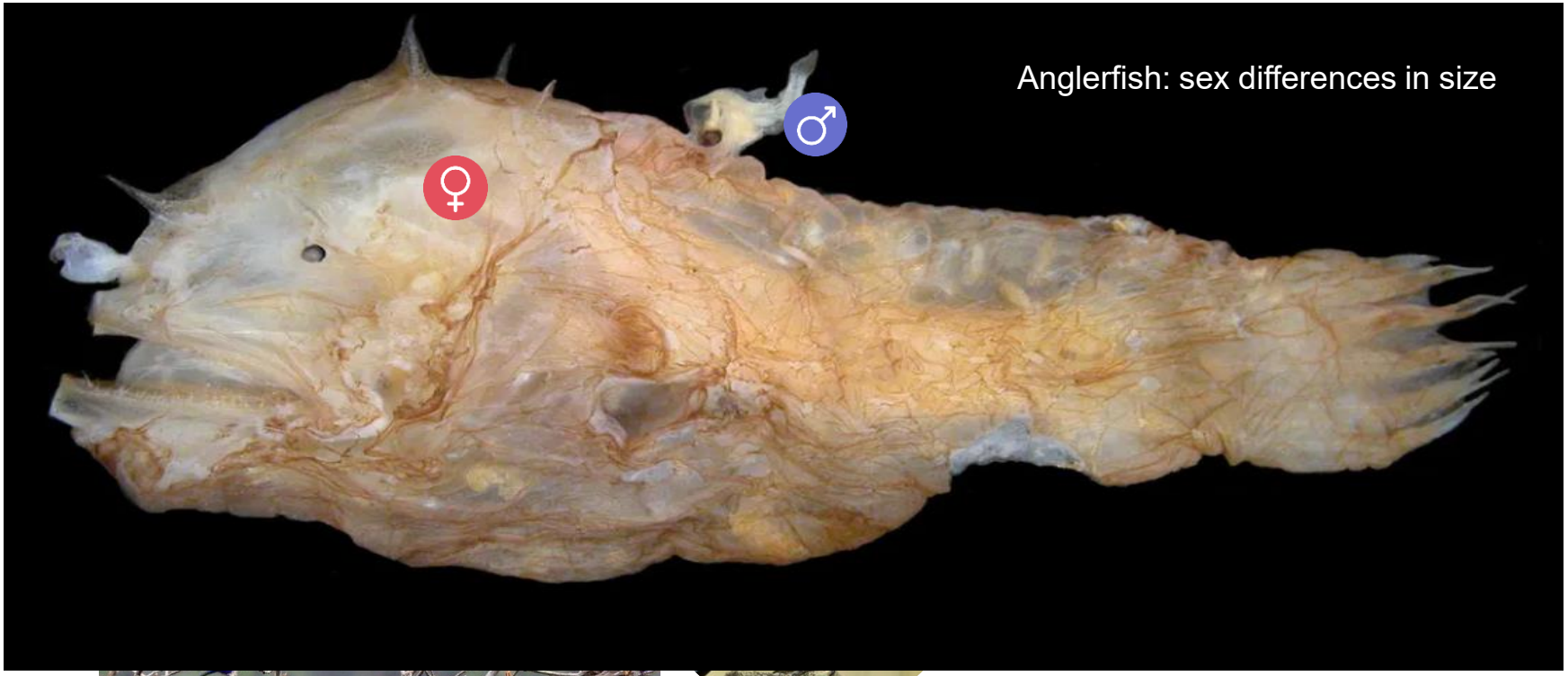


Female-typical



Vary with: age, experience, reproductive stage, health

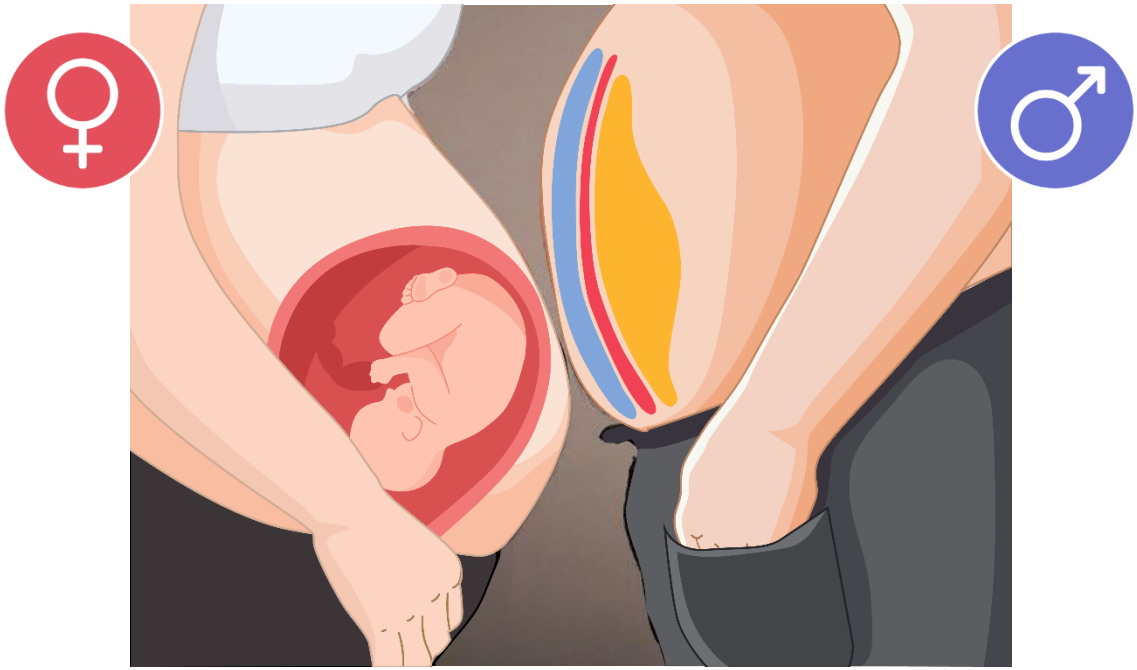
Sex differences in reproductive-related body morphology



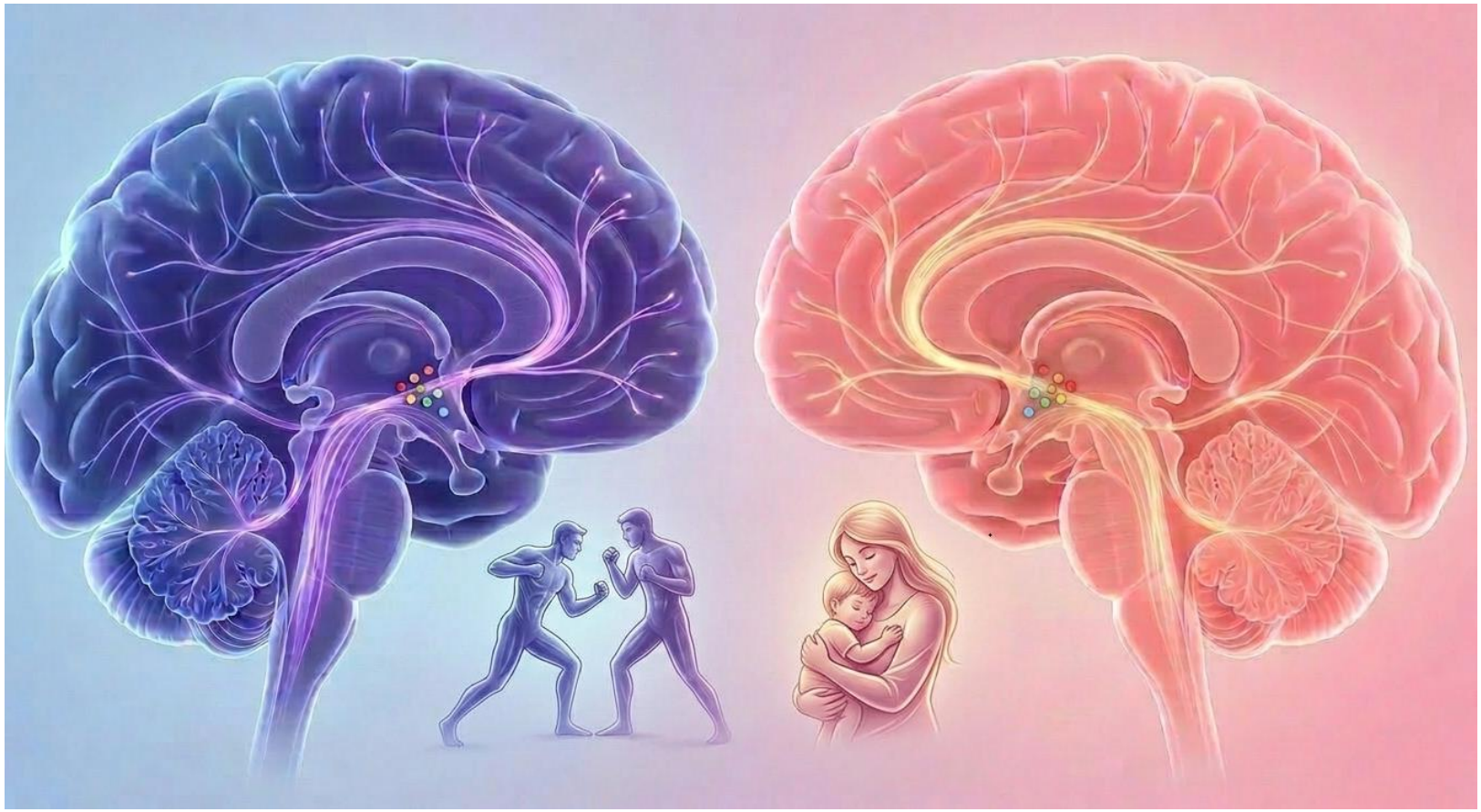


Rajesh Panwar

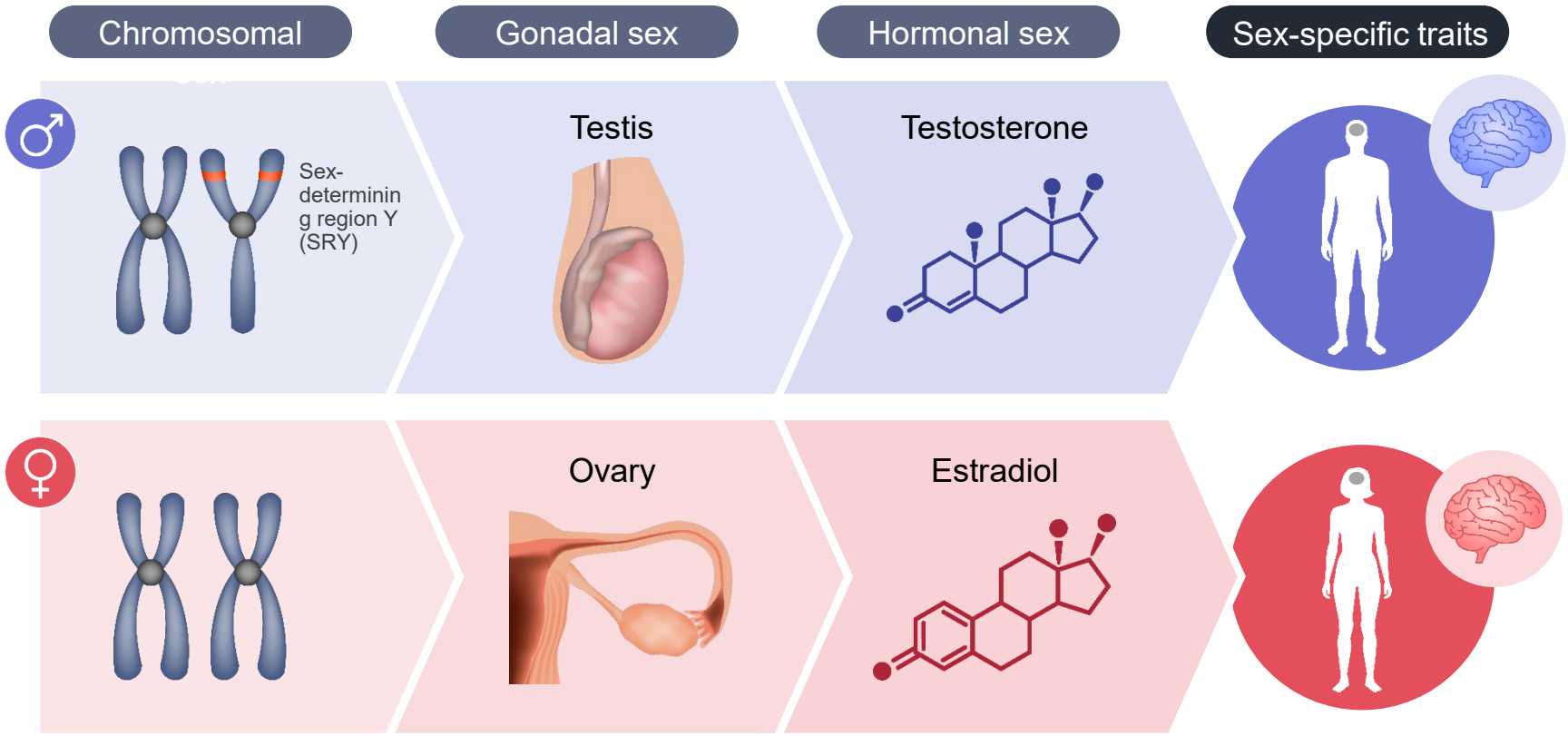
Sex-specific morphology



How do behavioral differences between females and males reflect in their brains?



Three ways to define “sex” biologically



Sex hormones and physiology



Behavioral Neuroendocrinology



Arnold A. Berthold
(1803-1861)



Hypothesis: Intact testes are necessary for the development of male-typical characteristics

Research question tested: Are the effects of prepubertal castration in males, dependent on neural connections to the testes?



Castration





Castration +
Transplantation
in the same body







Castration +
Transplantation
in another body



Berthold's experiment- summary

Findings:

-Males that were castrated as juveniles later showed deficits as adults, in male-typical morphology and behaviors.

-All of these effects could be reversed if the subject's testes, or the testes of another male, were implanted into the body cavity.



Conclusion:

Testes influence the development of male-typical morphology, and male-typical behavior, NOT through nerves, but by secreting a substance into the bloodstream (i.e. hormones).



William C. Young
(1899-1965)



Suggested that the same kind of relationship between embryonic and adult effects of hormones might hold in the brain.

Endocrinology, 1959

ORGANIZING ACTION OF PRENATALLY ADMINISTERED TESTOSTERONE PROPIONATE ON THE TISSUES MEDIATING MATING BEHAVIOR IN THE FEMALE GUINEA PIG[1]

CHARLES H. PHOENIX, ROBERT W. GOY, ARNOLD A. GERALL AND WILLIAM C. YOUNG

Department of Anatomy, University of Kansas, Lawrence, Kansas



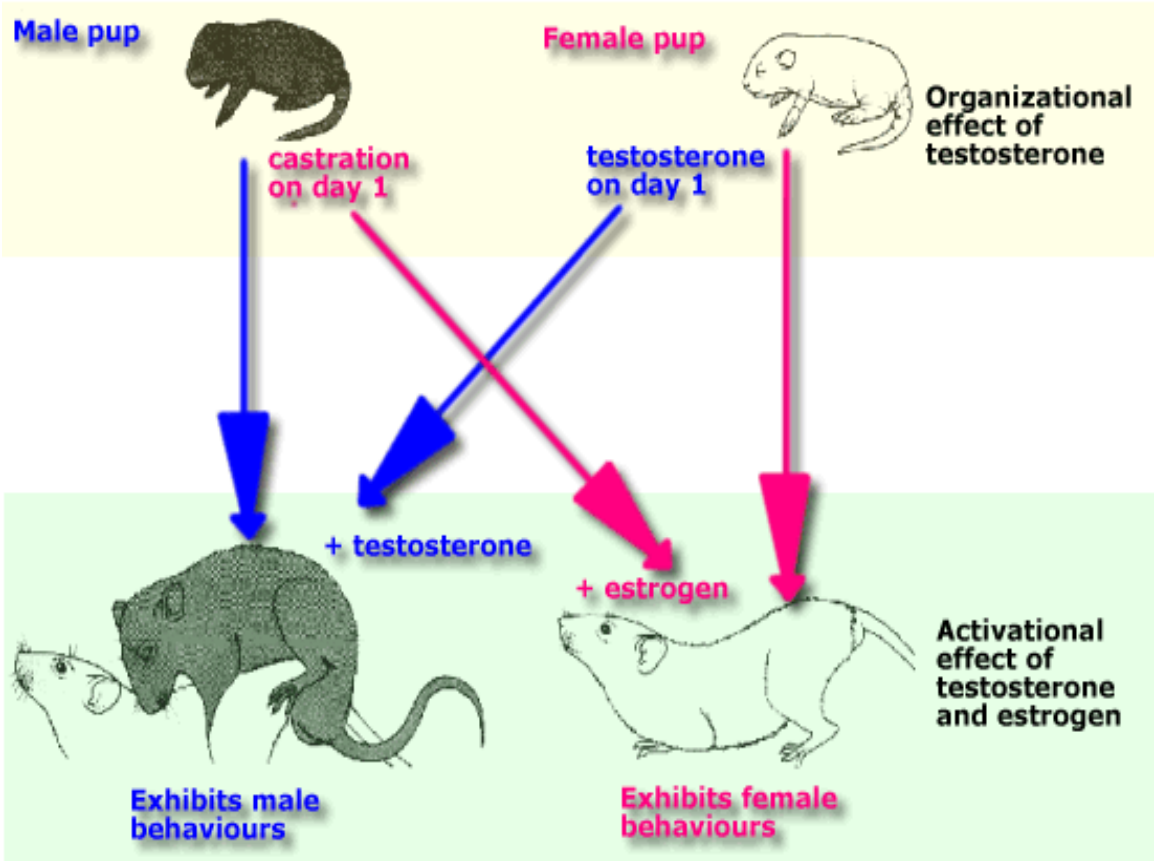
William C. Young
(1899-1965)

Female guinea pigs that had been exposed to androgens *in utero* were much more likely to express male-like copulatory behavior when exposed to androgens as adults.

They were also less likely to express female-typical receptive behavior when treated with estrogen and progesterone.

Endocrinology, 1959

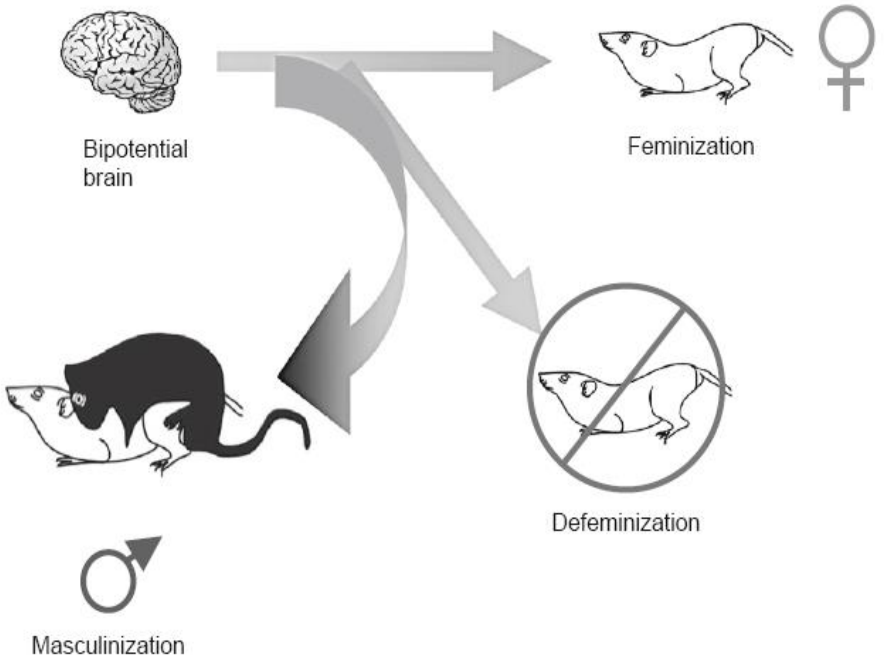
Effects of male castration and female testosterone treatment



The organization/activation hypothesize

•Sex hormones act during the prenatal stage to permanently (irreversibly) organize the nervous system in a sex-specific manner

•During adult life, the same hormones have activation effects, causing it to function in a sex-typical manner



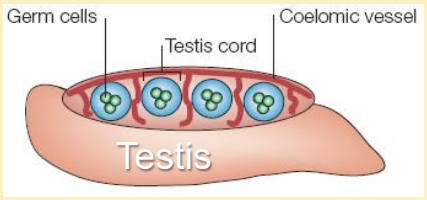
The classic model of brain sexual differentiation

Embryonic

XY ♂

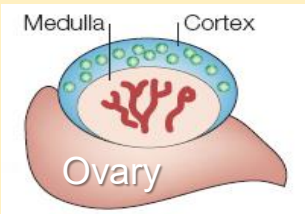
♀ XX





SRY gene

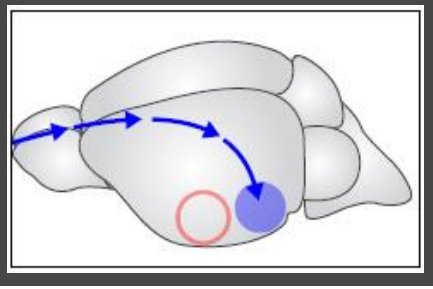
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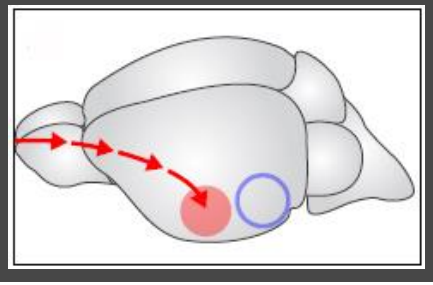


Perinatal

Testosterone/
Estradiol

Organization
(permanent changes)





Adult

Testosterone

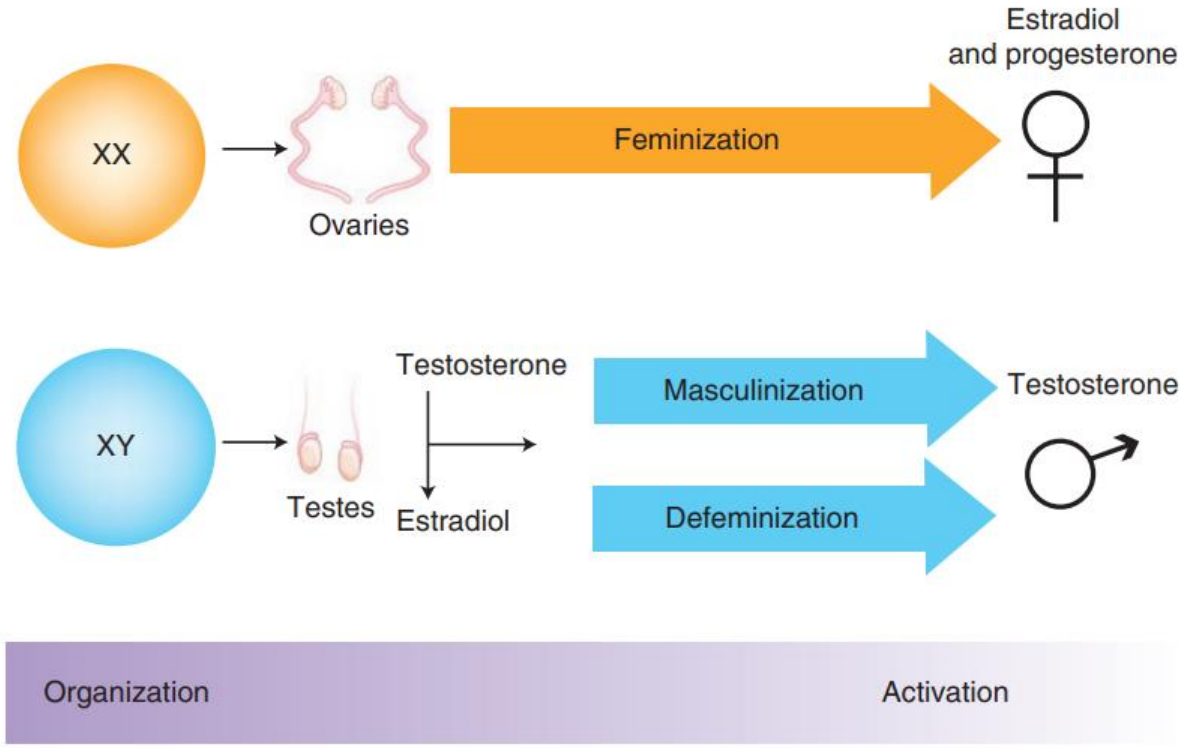
Activation

Estradiol &
Progesterone

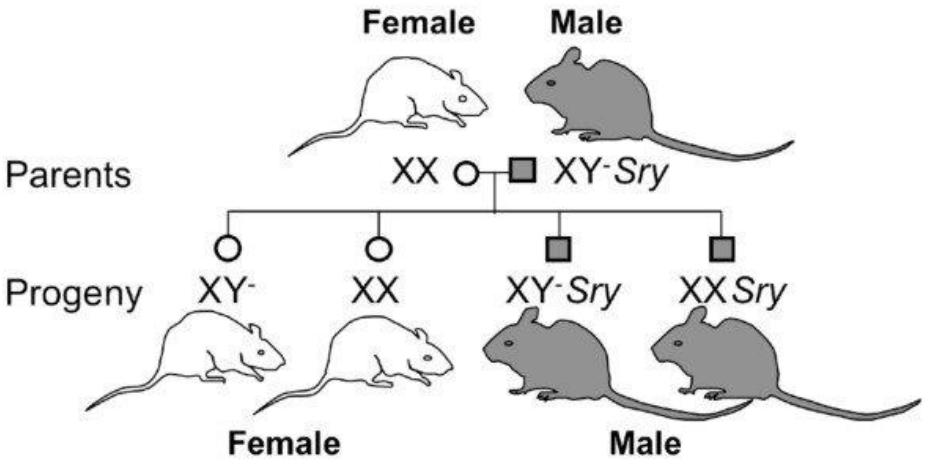


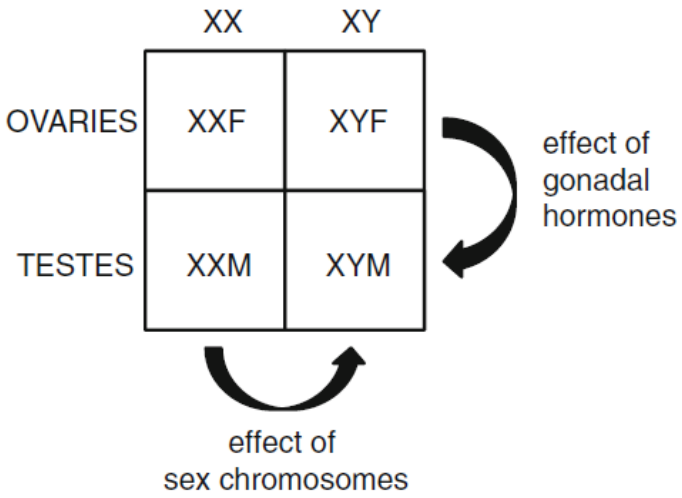


Classic model: hormone-mediated organization/activation



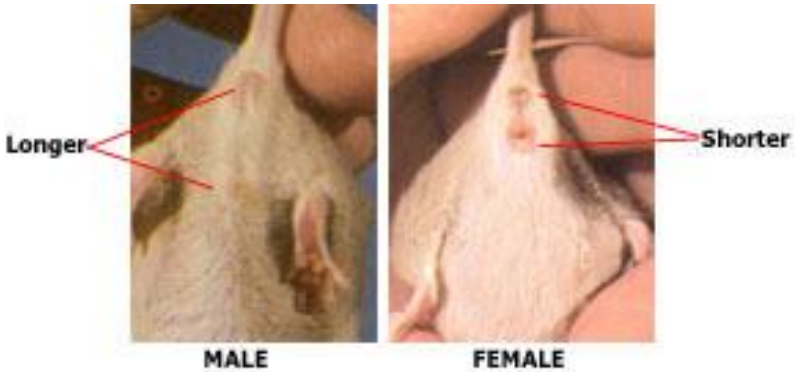
Genetic manipulation of Y chromosome and SRY gene in mice (Four Core Genotypes)

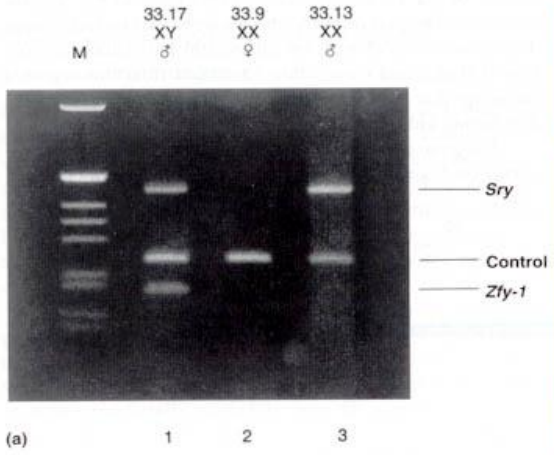




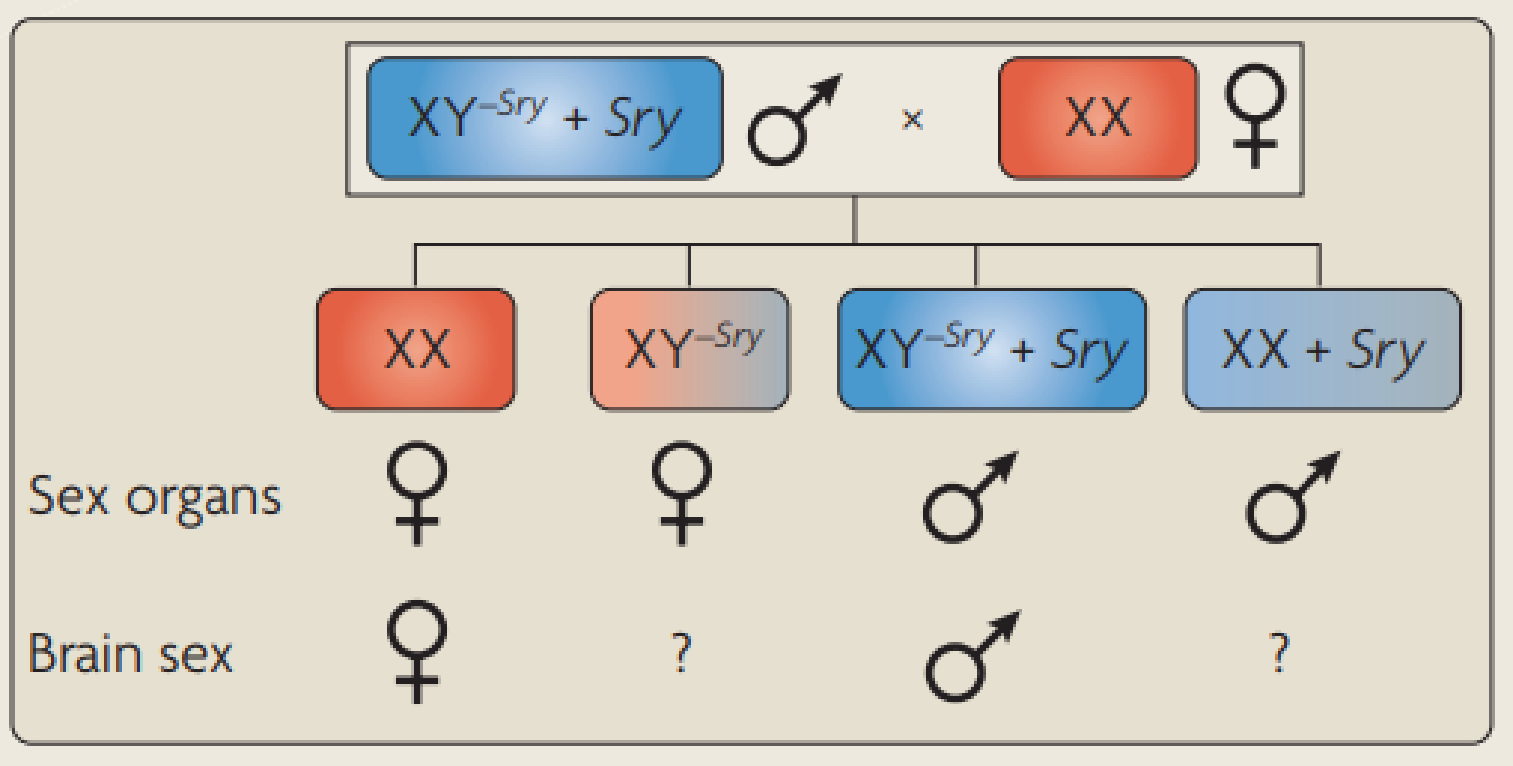
Burgoyne and Arnold, 2016;
Burgoyne et al., 1998;
De Vries et al., 2002

Genetically Female Mice Transgenic for SRY are Phenotypically Males

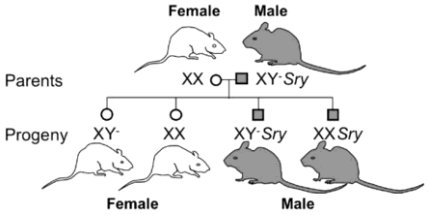




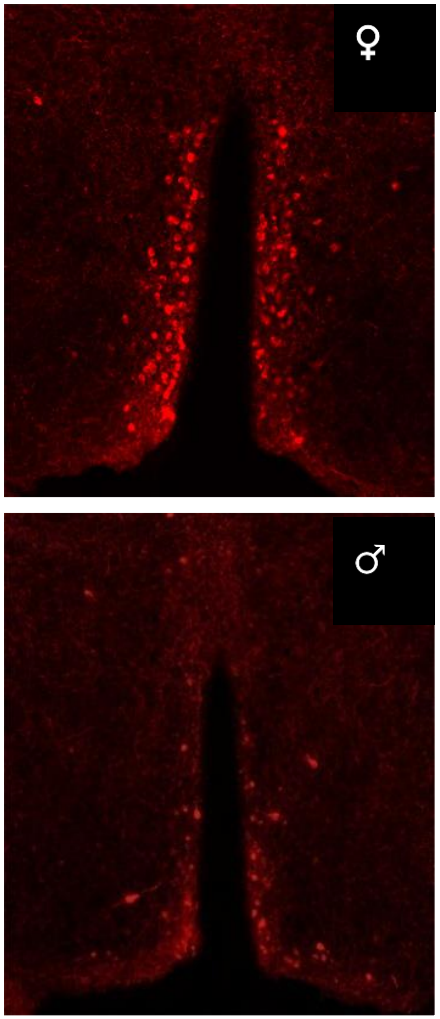


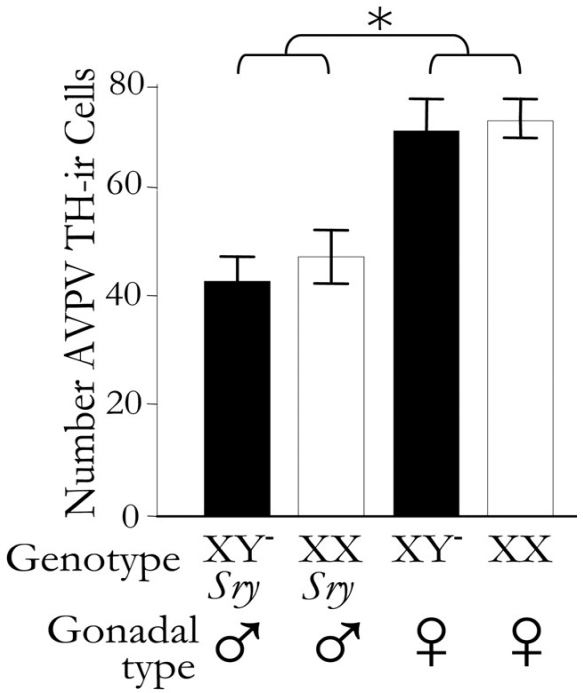

XY−Sry + Sry
×
XX
XX
XY−Sry
XY−Sry + Sry
XX + Sry
Sex organs
Brain sex
?
?

Sry controls the expression of sexually dimorphic gene (TH) in the brain



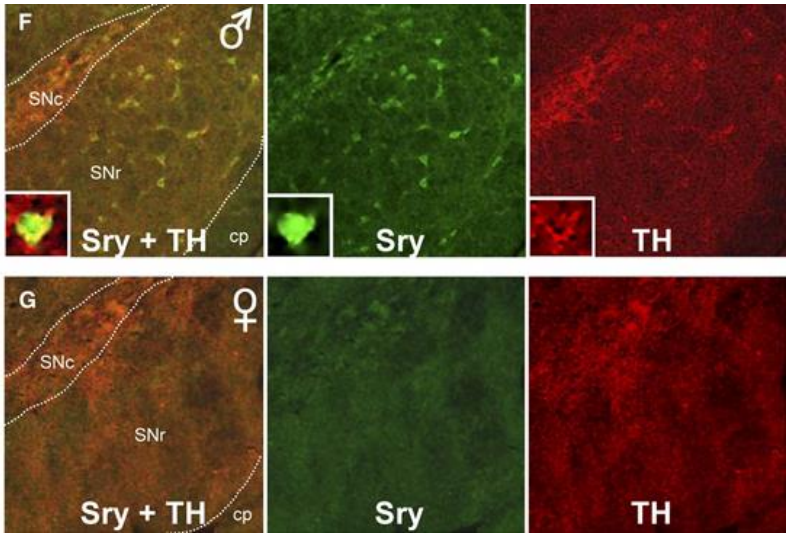
Tyrosine Hydroxylase (TH)-expressing neurons in the hypothalamus

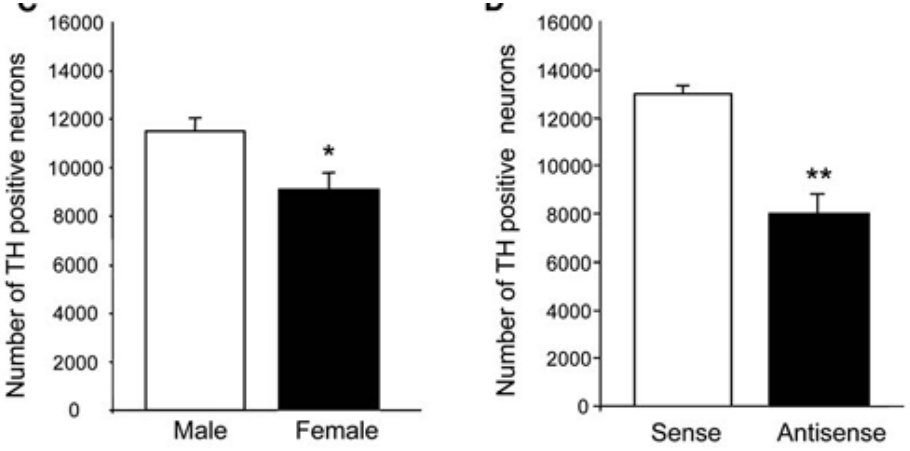




Arnold et al 2004, Endocrinology

SRY expresses in the male brain and regulate TH expression

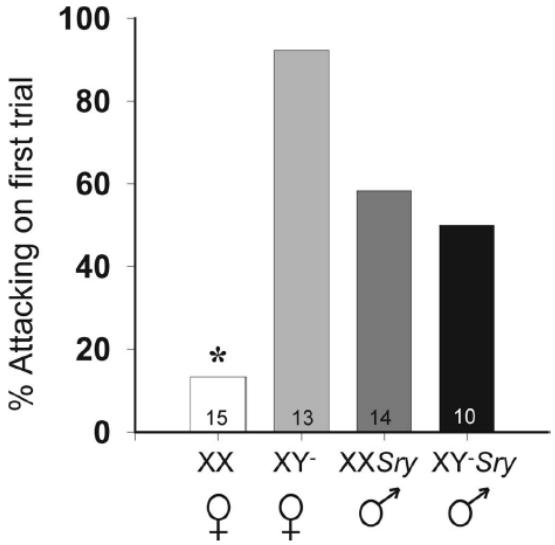




Dewing et al 2006

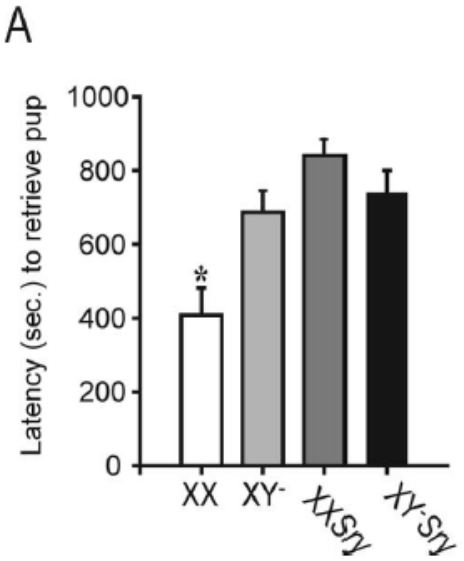
Weak evidence that SRY regulates sex-typical social behavior

Aggressive behavior



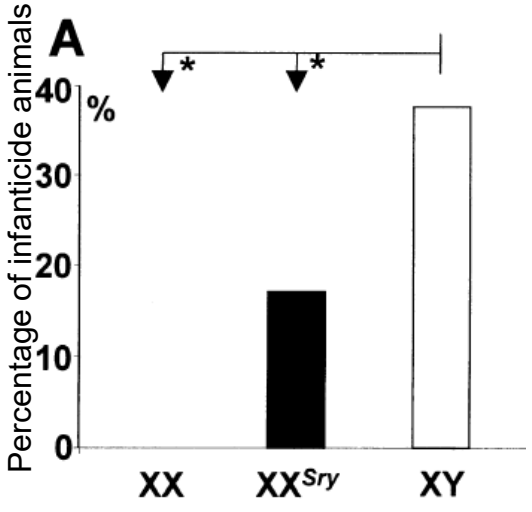
Gatewood et al 2006, J Neuros

Maternal behavior



Gatewood et al 2006, J Neuros

Infanticide behavior



Ingrid et al 2002, Behav Genet

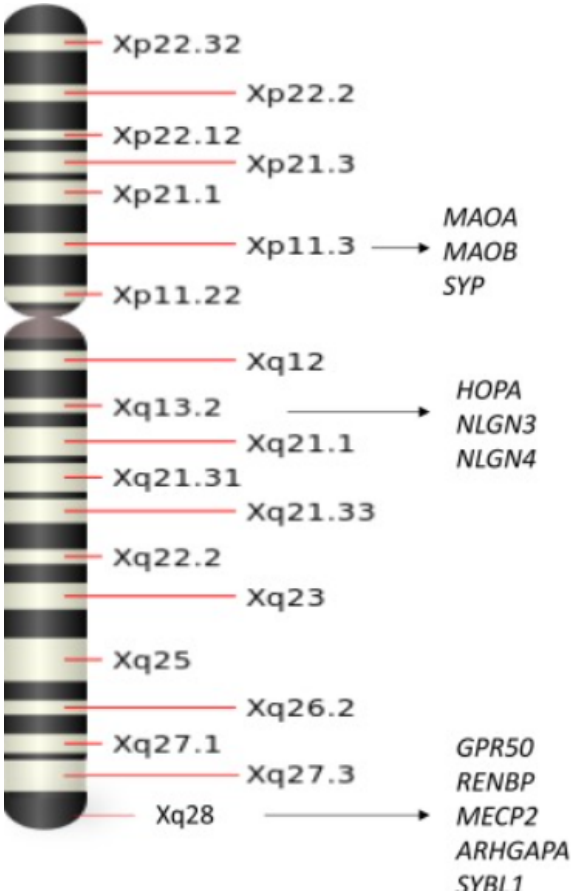
The Klinefelter syndrome, also known as the **XXY genetic disorder**

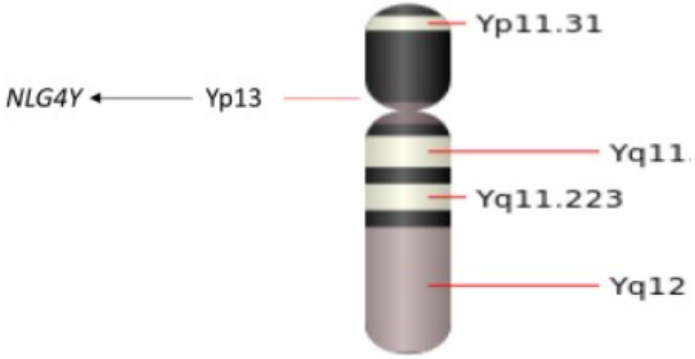


symptoms include:

1. Reduced fertility or full infertility
2. Female-typical body characteristics

A schematic of the sex chromosomes with several genes associated with psychiatric diseases





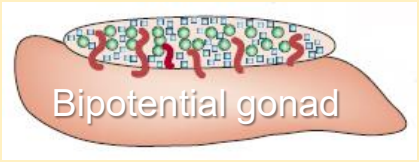
X chromosome: Contains ~ 900 to 1,200 protein-coding genes.
Y chromosome: Contains only ~50 to 200

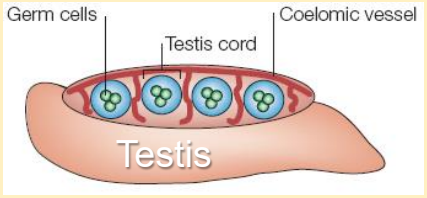
The classic model of brain sexual differentiation

Embryonic

XY ♂

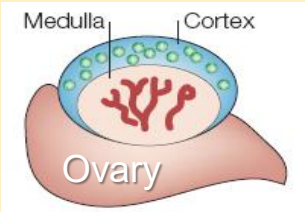
♀ XX





SRY gene

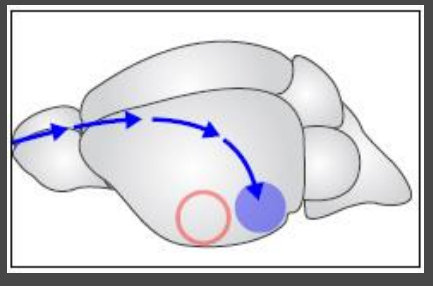
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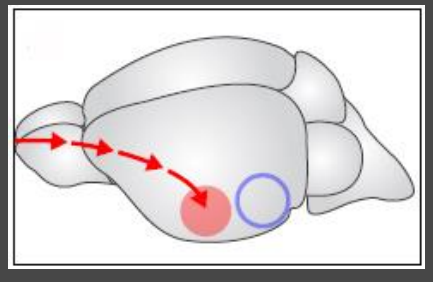


Perinatal

Testosterone/
Estradiol

Organization
(permanent changes)





Adult

Testosterone

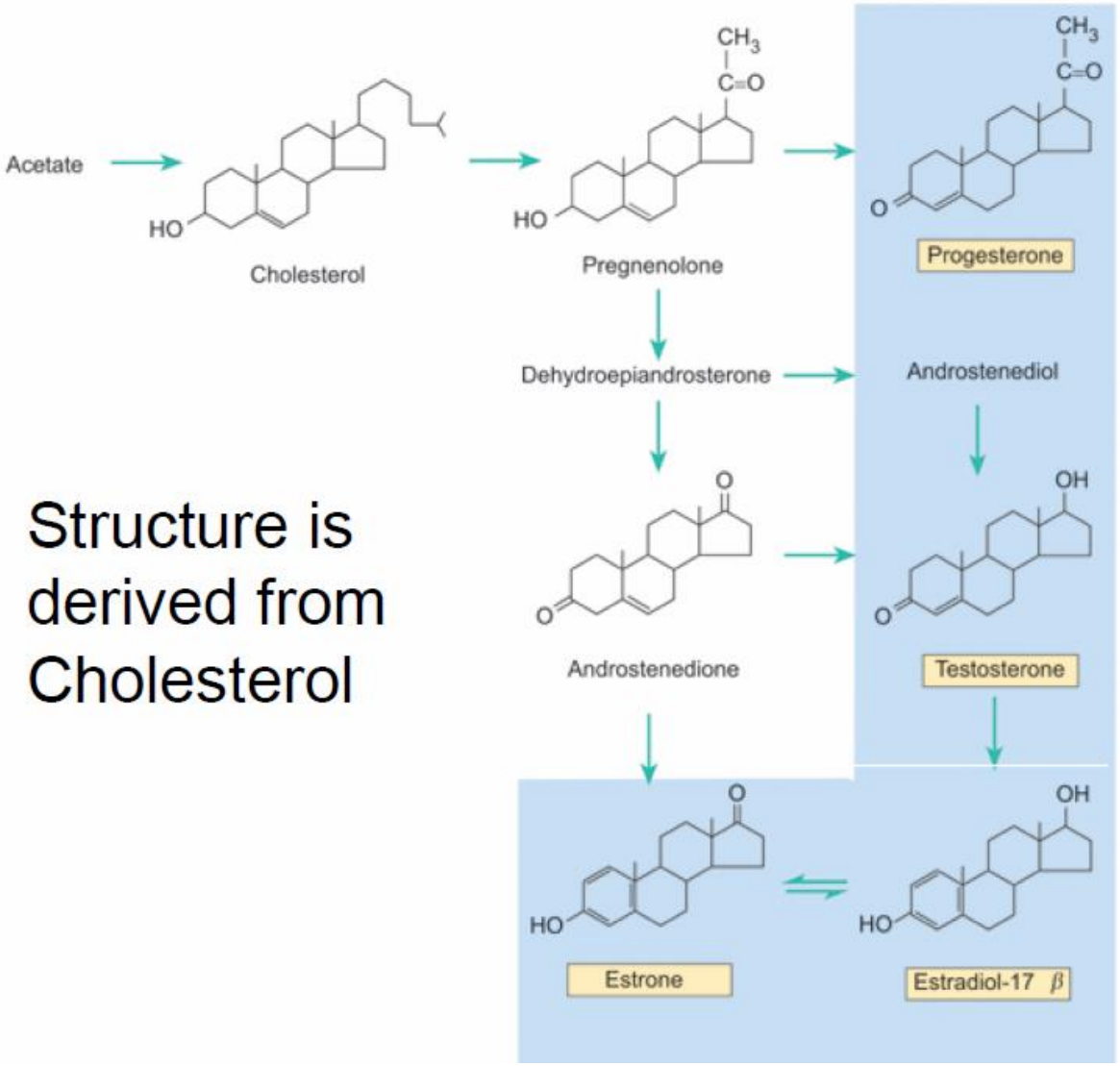
Activation

Estradiol &
Progesterone





The sex hormones



Structure is derived from Cholesterol

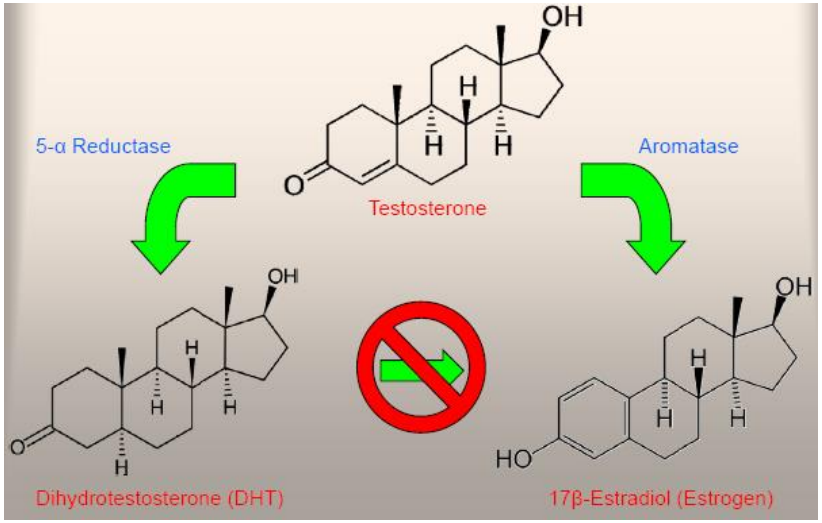
"Male" (androgenic) sex hormone

-Also secreted by the adrenal gland

"Female" (estrogenic) sex hormone

* Both are steroid hormones and secreted in both sexes

Testosterone, Estradiol or DHT masculinizes the brain ?



- Testosterone treatment in neonatal rats is blocked by prior administration of specific estrogen receptor antagonist

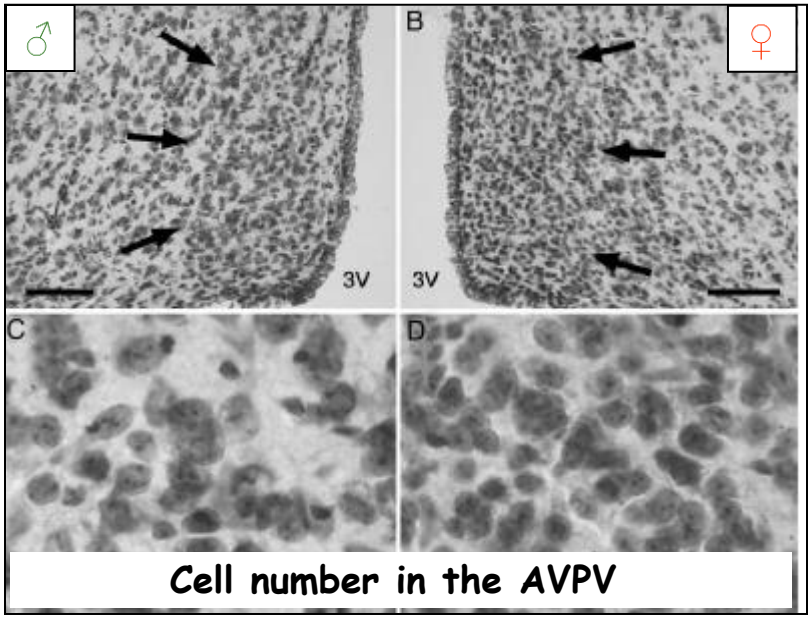
- DHT does not mimic the effect of testosterone

- Radio-labeled testosterone is recovered from the brain as radio-labeled estradiol

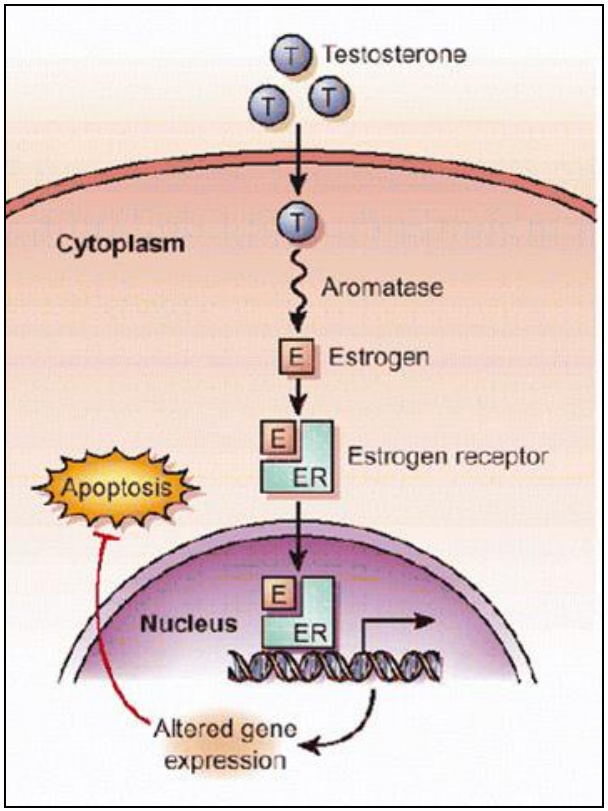
- Aromatase inhibitors counteract the effect of testosterone administration

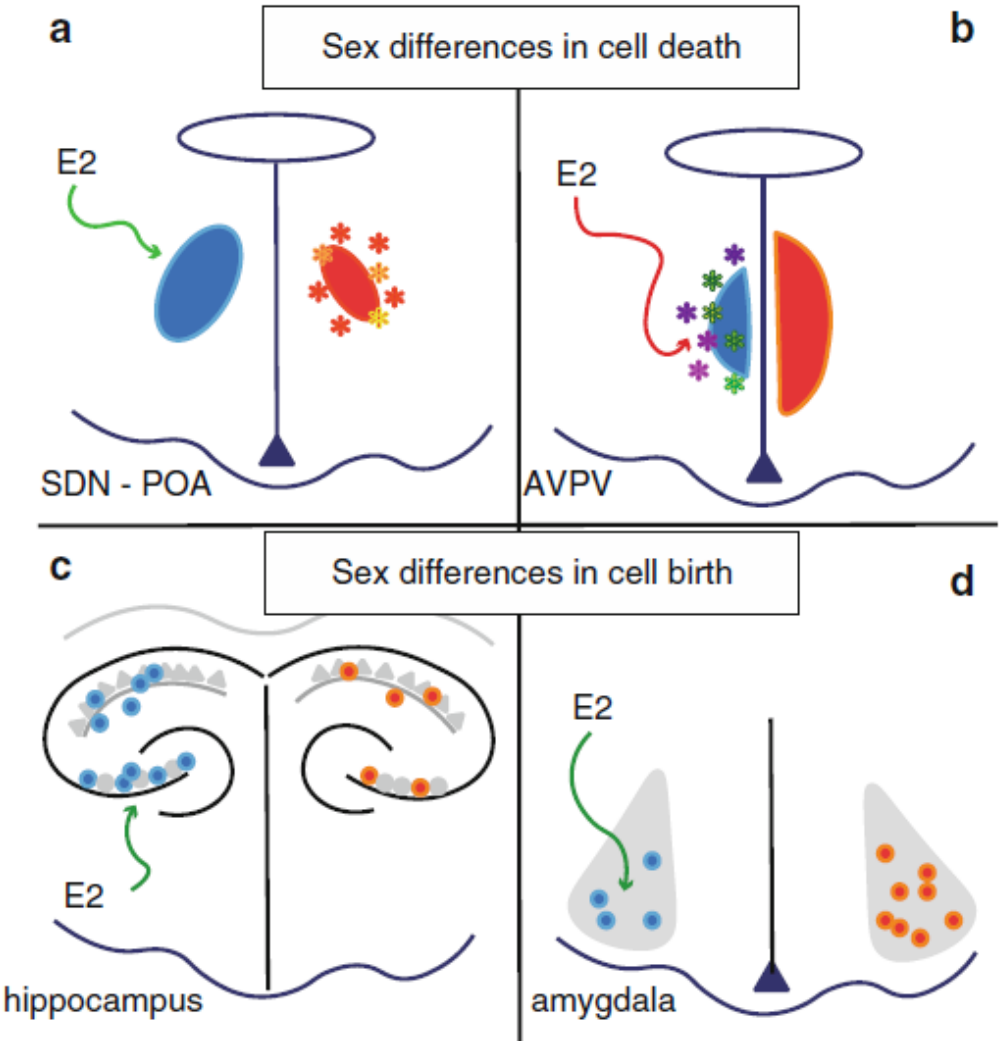
Estradiol masculinizes the brain

Cell death and a sexually dimorphic brain nucleus (AVPV)



Cell number in the AVPV





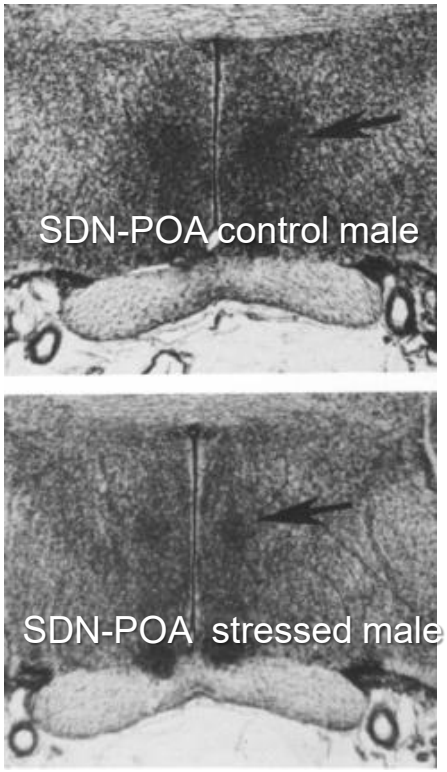
Arnold and McCarthy, 2016

Epigenetic mechanisms underlying the development of sex-typical brain and behavior

Effect of prenatal stress on sexual dimorphism in the rat brain

SDN measurements

Treatment	*Days postnatally*		
	Birth	*20 Days*	*60 Days*
Males			
Control	0.259 ± 0.015	**0.858 ± 0.083	**0.643 ± 0.035
ES	*0.471 ± 0.034	*0.419 ± 0.049	*0.345 ± 0.034
NS	*0.447 ± 0.027	*0.553 = 0.086	*0.278 ± 0.039
Females			
Control	0.261 ± 0.021	0.369 ± 0.027	0.378 ± 0.025
ES	0.324 ± 0.034	0.440 ± 0.076	0.258 ± 0.023
NS	0.363 ± 0.033	0.502 ± 0.066	0.339 ± 0.036

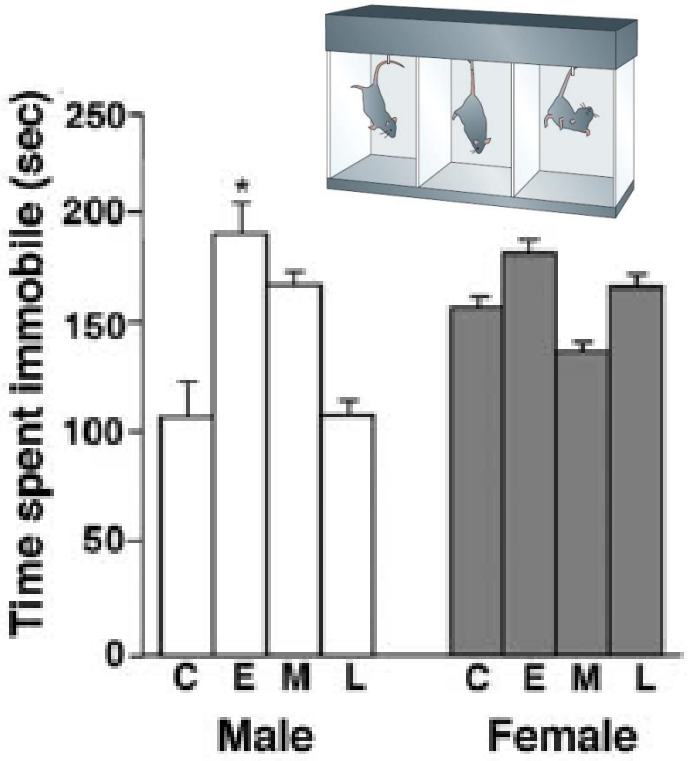


ES: Environment stress (change in lighting/ temperature)
NS: nutritional stress (50% of total food of control males)

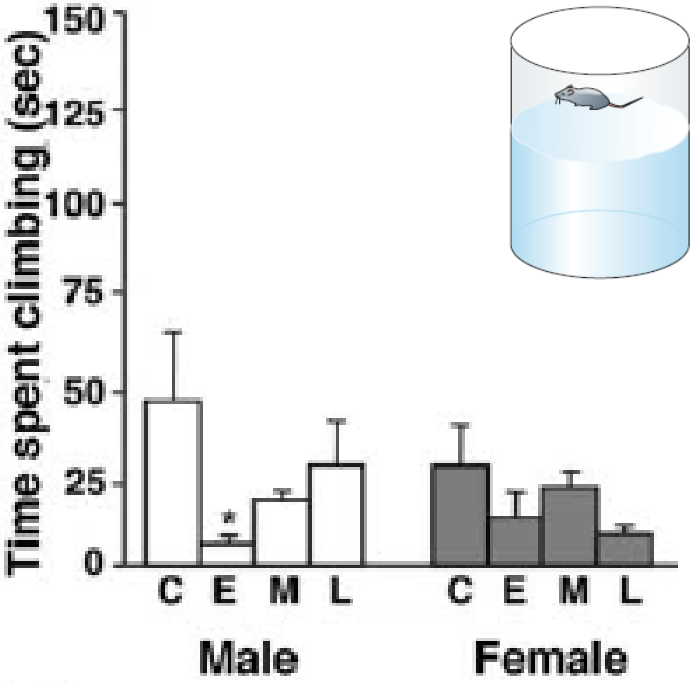
Anderson et al 1985

Sex-specific programing of offspring behavior after stress early in pregnancy

Tail suspension assay



Forced swim test



C: control;
Prenatal stress during (E) early, (M) mid or (L) late gestation

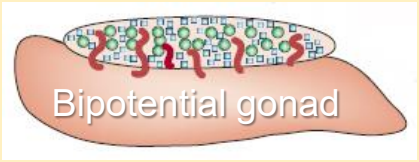
Mueller and Bale 2008

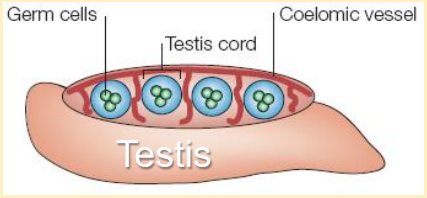
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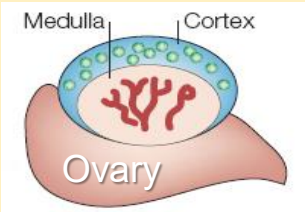
♀ XX





SRY gene

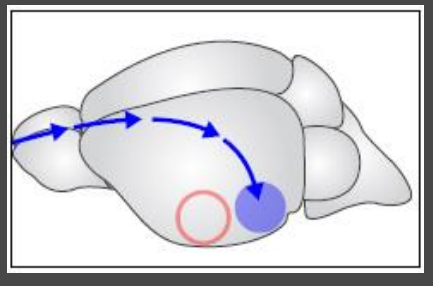
\+ -

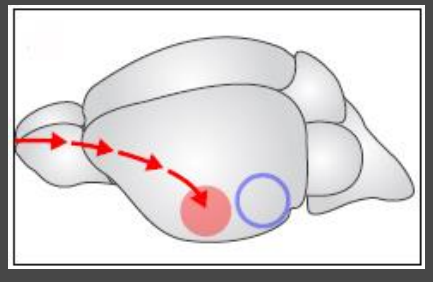


Perinatal

Testosterone/
Estradiol

Organization
(permanent changes)





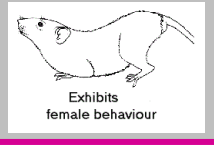
Adult

Testosterone

Activation

Estradiol &
Progesterone

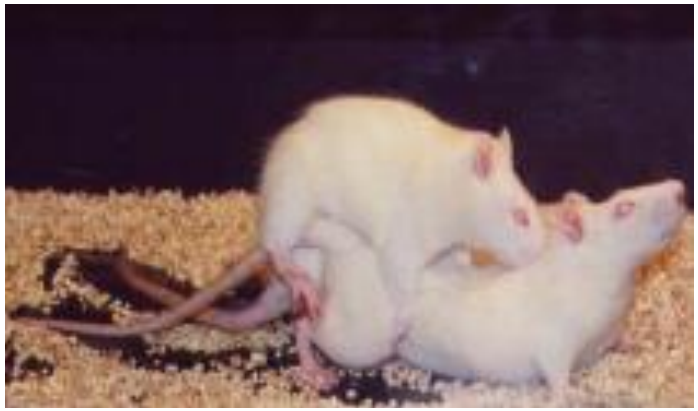




Hormonal regulation of social behavior during adulthood



Aggressive behavior



Sexual behavior



Maternal behavior

Maternal behavior in postpartum female rats



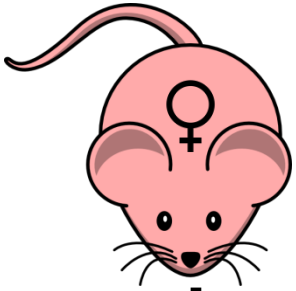






Pup-directed behavior in virgin and postpartum female

Virgins





Infanticide

Postpartum





Maternal

Measuring parental behaviors in mice

Pup nursing behavior (retrieval) by females

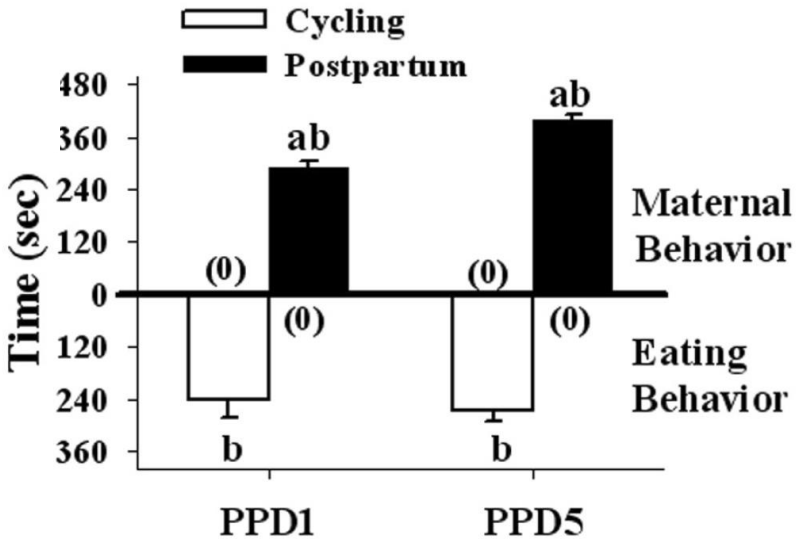


Pup-directed behavior (ignoring) by males

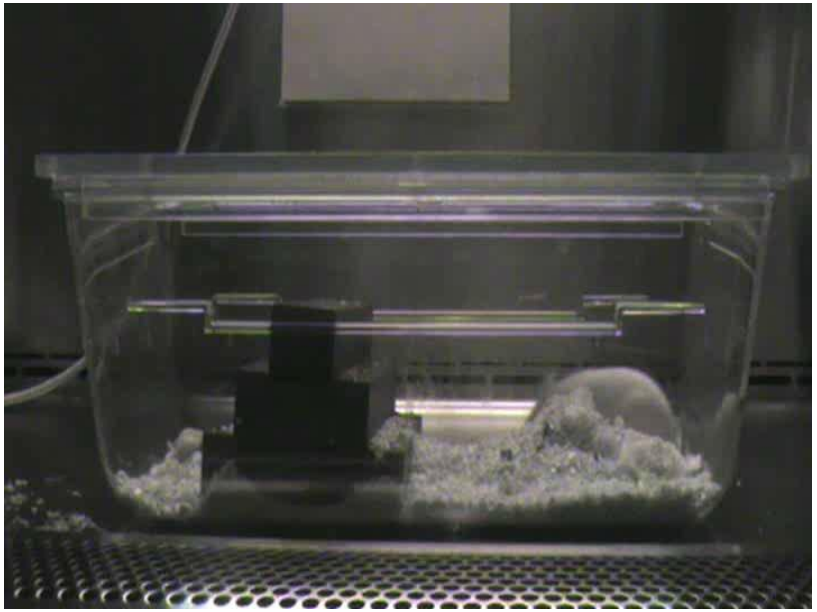


Pups are highly motivated stimuli for dams





Afonso et al 2009



Scott et al 2015

Comparison of Two Positive Reinforcing Stimuli: Pups and Cocaine Throughout the Postpartum Period

B. J. Mattson, S. Williams, J. S. Rosenblatt, and J. I. Morrell
Rutgers, The State University of New Jersey

Behav. Neurosci. 2001

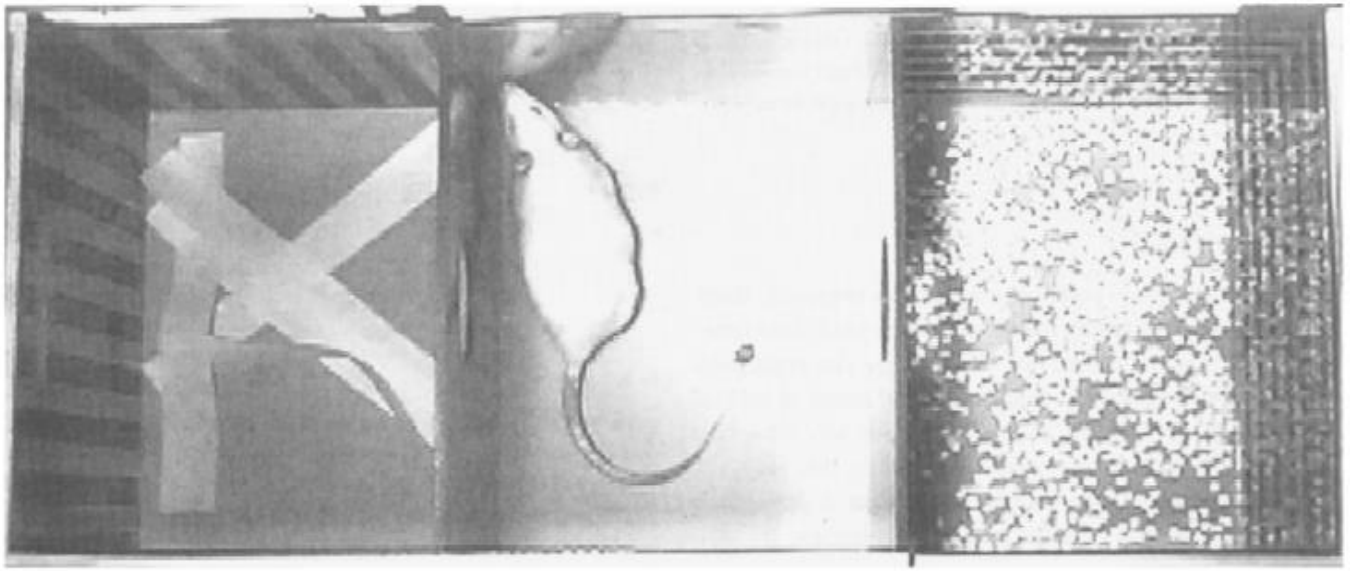


Figure 1. Place preference paradigm apparatus as prepared for a typical testing session. The tactile (strips and squares) and visual (vertical and horizontal) contextual cues in the outer chambers are visible. Inner walls are red, with a 10.1-cm hole, which enabled the dam to travel throughout the apparatus.

Studying maternal behavior motivation using conditioned place preference (CPP) test

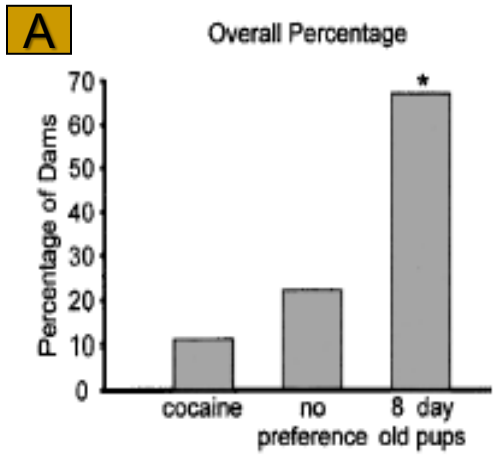
- The CPP procedure assesses the preference for or the motivation to seek a reinforcing stimulus, including a variety of natural reinforces.
- Animals are given pairings of an unconditioned reinforcing stimulus with a set of unique environmental cues that serve as the conditioned stimulus.



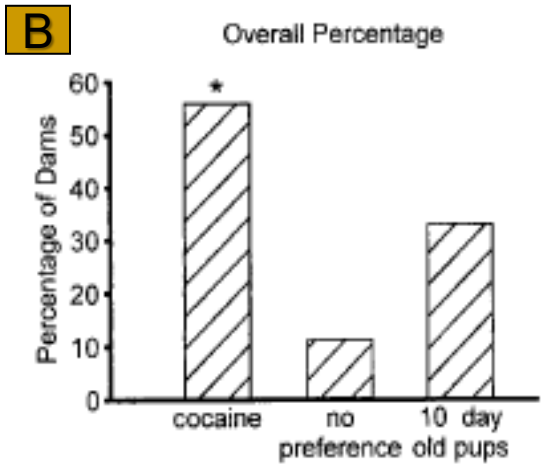
The reward value of pup stimuli to mothers

- Females were exposed to unconditioned stimuli (pups or cocaine) in the presence of conditioned stimuli cues.
- On postpartum day 8 (**A**), 10 (**B**) or 16 (**C**), the # of dams spent in each chamber and their behavior were recorded.

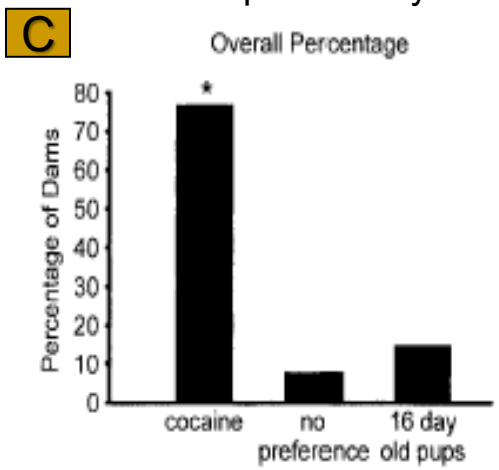
Postpartum day 8



Postpartum day 10



Postpartum day 16

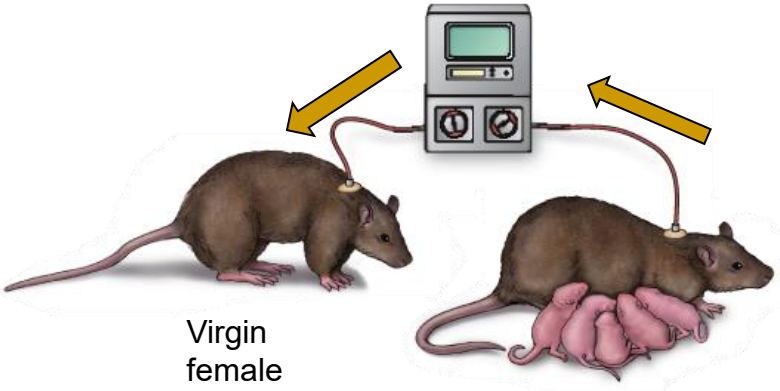


Mattson et al. 2001

Maternal behavior induced by maternal blood plasma



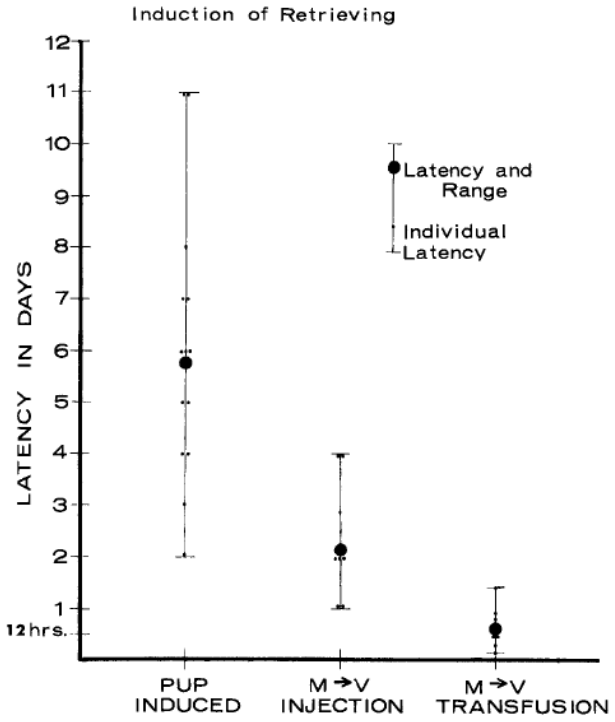
Joseph Terkel



Blood was transfused from a parturient female (one that had given birth within 30 min prior to the onset of the transfusion) into a virgin female.

Terkel and Rosenblatt, 1968

Hormonal factors underlying maternal behavior



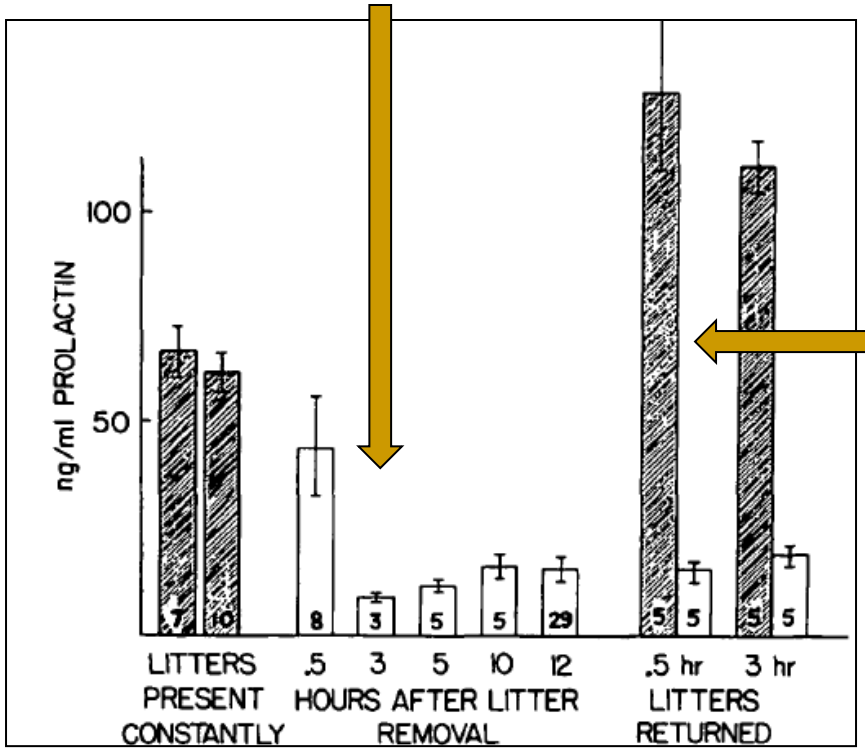
Maternal behavior of virgin females toward newborn (unfamiliar) pups is facilitated following blood transfusion from maternal females (lactating)

Terkel and Rosenblatt, 1972

Prolactin in serum and maternal care in rats



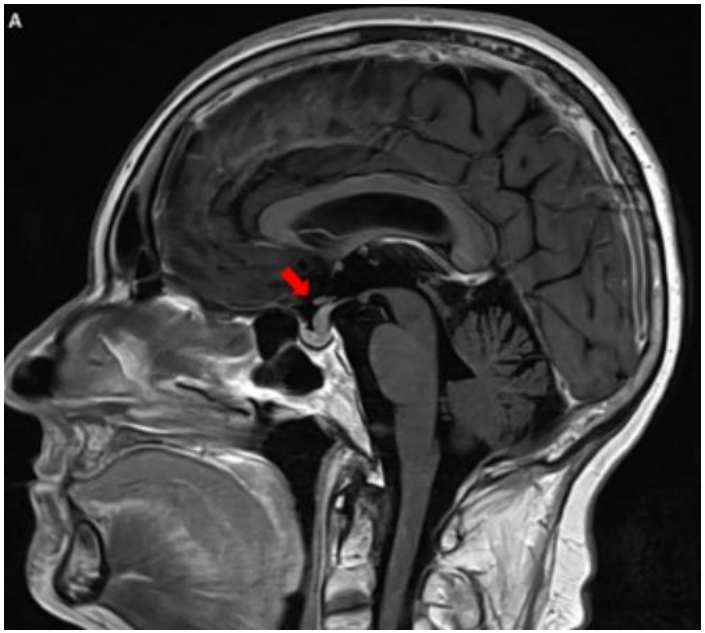
• Removal of litters from mother rats resulted in a rapid decline of serum prolactin levels, reaching pregnancy levels 3 hr later

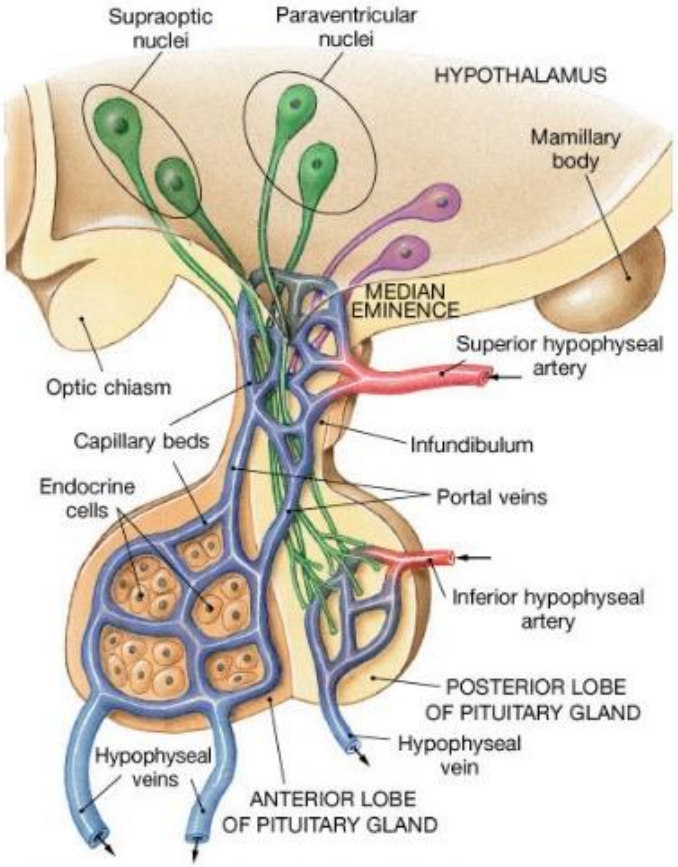


• When litters were returned to their mothers for 0.5-3 hr serum prolactin increased sharply

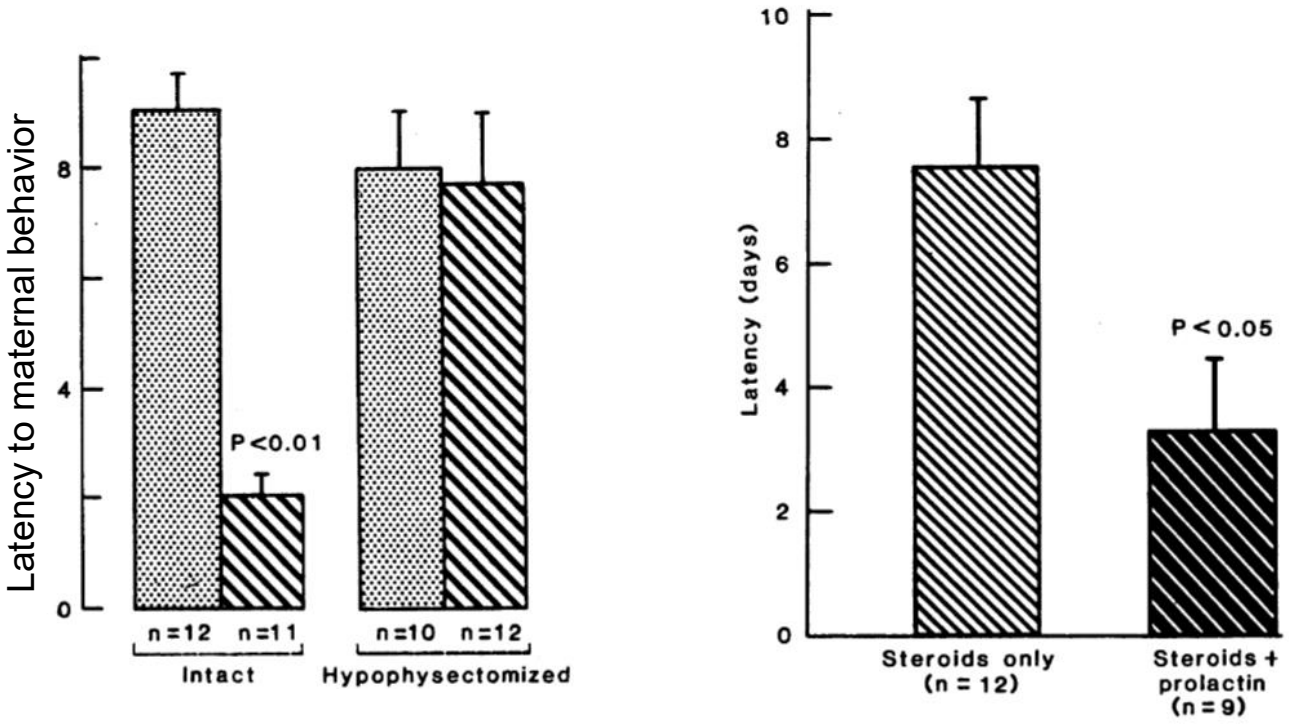
Amenomori et al 1970

Prolactin is released from the (anterior) pituitary gland and required for maternal care





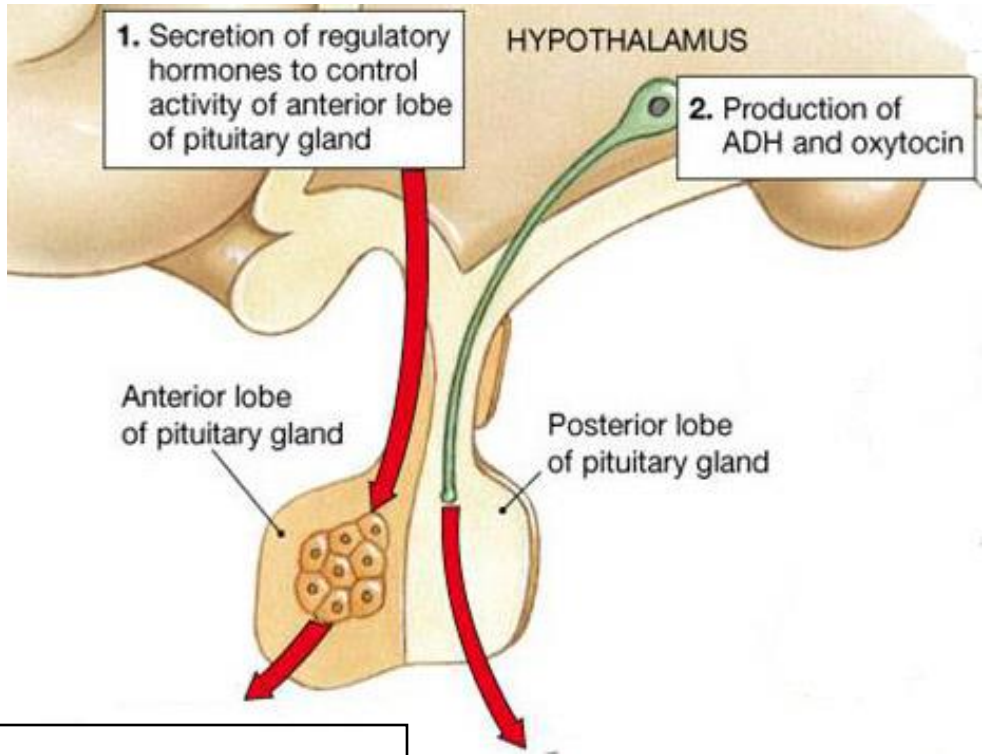
Prolactin is released from the pituitary gland and required for maternal care



*Naïve rats ovariectomized treated
with steroids (estradiol+progesterone)

Bridges et al. 1984

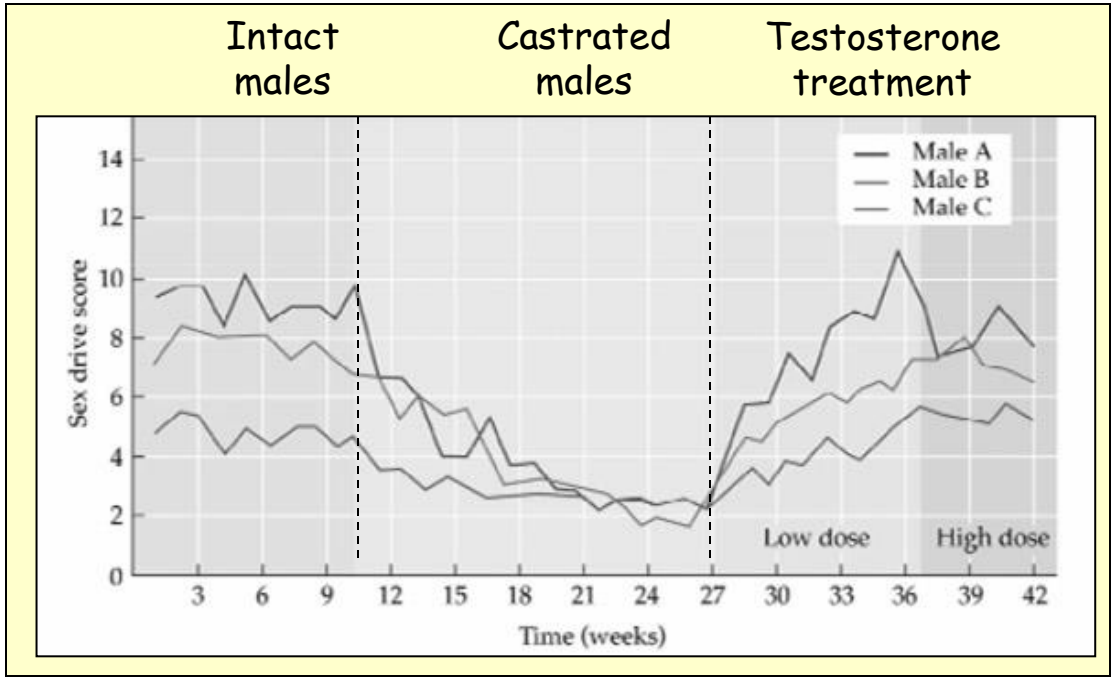
The Hypothalamus-Pituitary-Gonadal Axis



Prolactin (PRL)
Growth Hormone (GH)
Adrenocorticotropic hormone (ACTH)
Thyroid-stimulating hormone (TSH)
Follicle-stimulating hormone (FSH)
Luteinizing hormone (LH)

Vasopression
Oxytocin

Effects of castration & testosterone treatment on males



•In all rodents, gonadectomy decreases (abolishes) male courtship and sexual behavior

•Testosterone replacement reinstates sexual behavior in males

How can the female and male brains explain why females and males are so different?

Dimorphic brain functions/structures → Dimorphic social behaviors ?

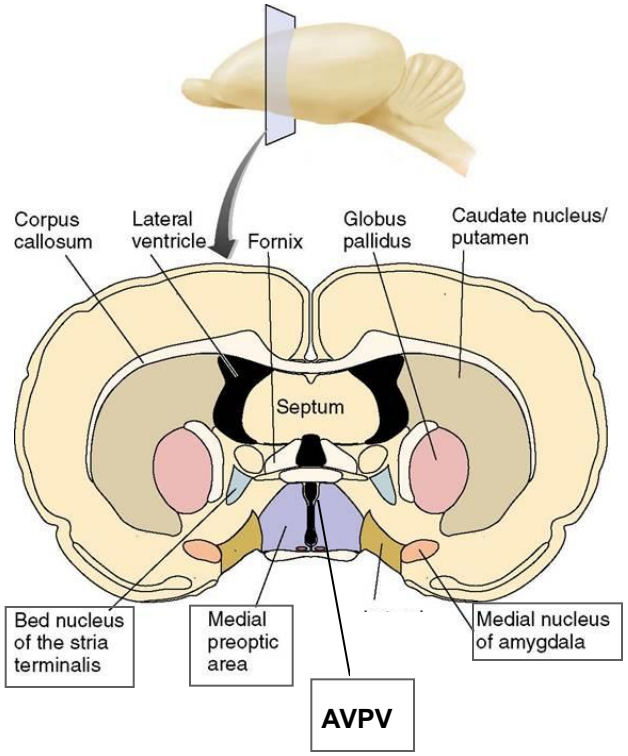
Sexually dimorphic brain nuclei in rodents

Bed Nucleus of the Stria Terminalis (BNST)
Sexually Dimorphic-Nucleus of Preoptic Area (SDN-POA)
Posterodorsal Medial Amygdala (MePD)

} Larger in male

Anteroventral Periventricular Nucleus (AVPV)

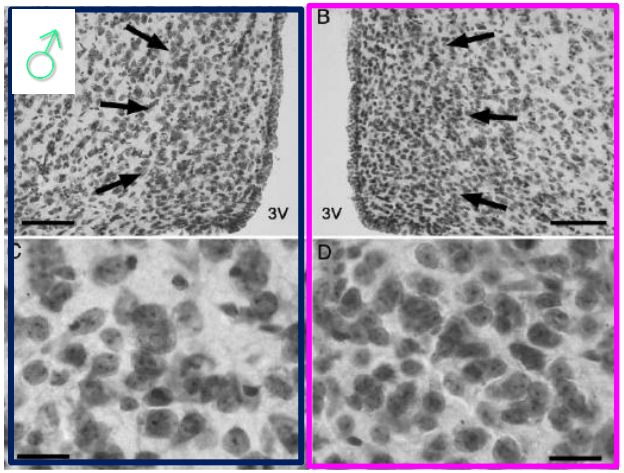
} Larger in Female



Region	Volume larger in
AOB	Male
MeA	Male
PMv	Male
Posterior BNST	Male
Anterior BNST	Female
Central nucleus	Male
SDN-POA	Male
AVPV	Female
SON	Male
SCN	Male
VMN	Male
Locus Ceroleus	Female
SNB	Male
POA	Male
MePD	Male

Adapted from, Wilson & Davis, *Reproduction* 2007; Forger et al. 2015

AVPV (F>M)



Forger et al 2004, *PNAS*

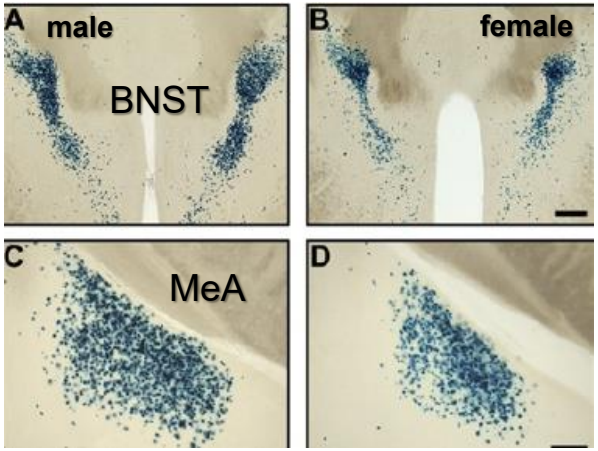
SDN-POA (M>F)



Gorski et al 1978, *Brain Res.*

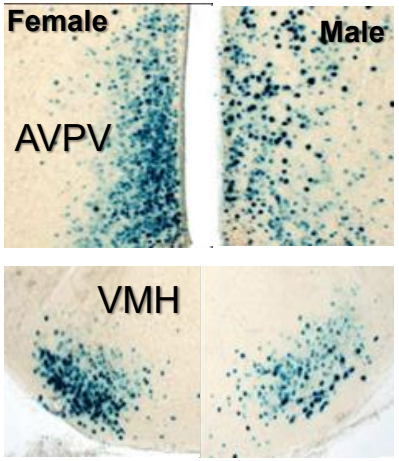
Sexual dimorphism: Gene expression

Androgen receptor



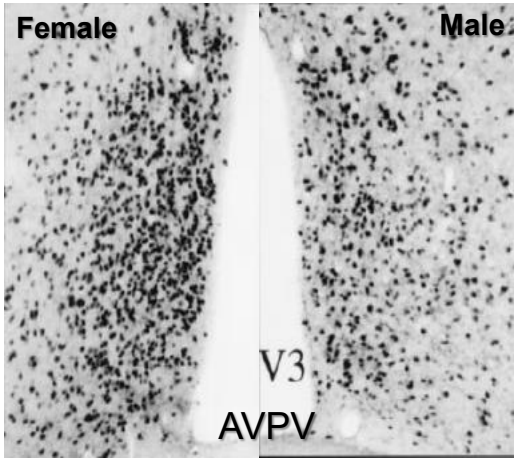
Juntti et al 2010, *Neuron*

Progesterone receptor



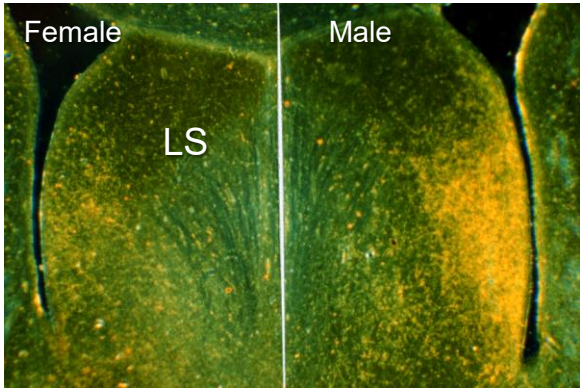
Yang et al 2013, *Cell*

Estrogen receptor α



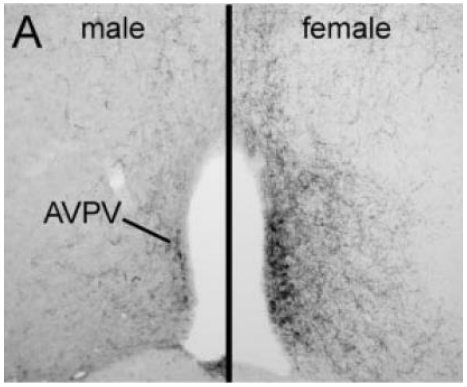
Simerly et al 1997, *PNAS*

Vasopressin fibers



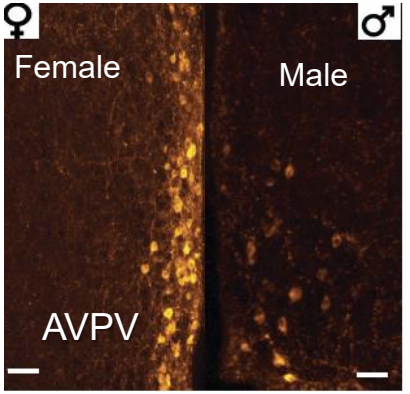
(Curtesy Geert de Vries)

Kisspeptin



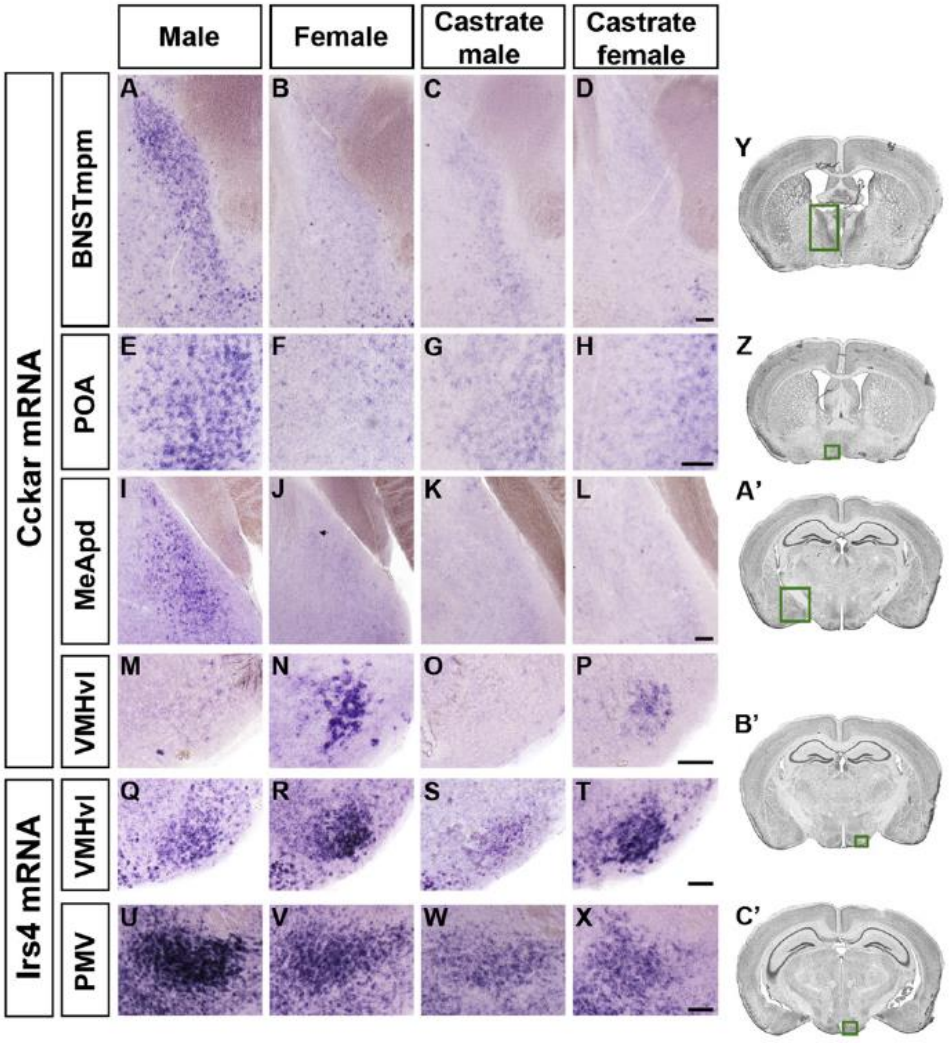
Clarkson & Herbison 2006, *Endocrinology*

Tyrosine Hydroxylase

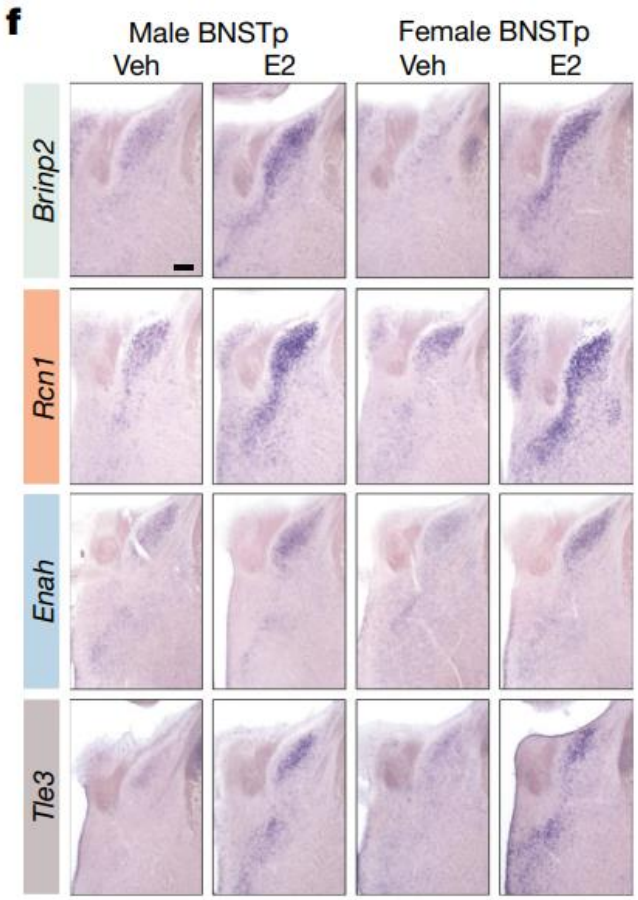


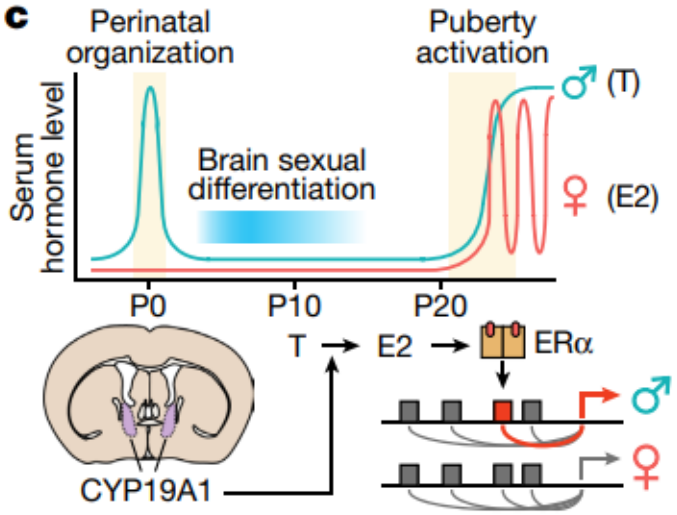
Scott et al 2015, *Nature*

Sex hormones regulate brain gene expression patterns in a sex-specific manner



ERα is required for sustained sex differences in gene expression

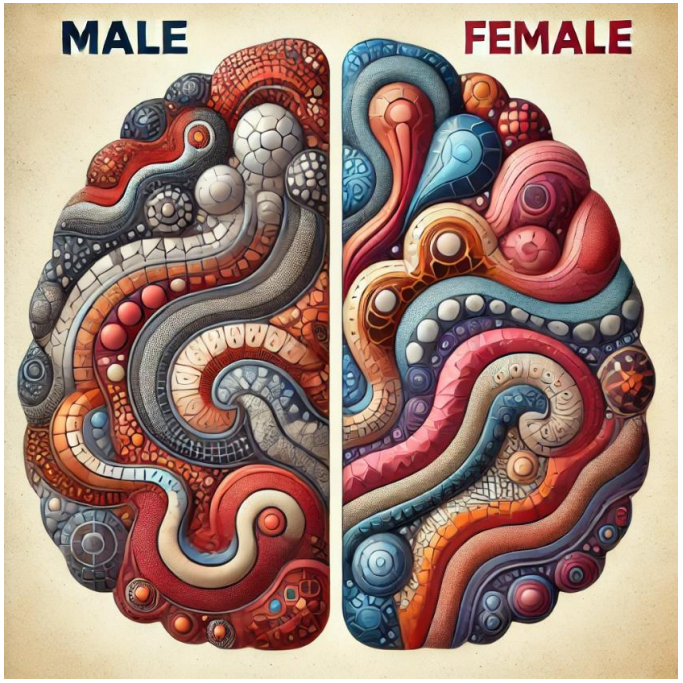




Neonatal hormone surge activates ERα to drive a male-biased gene expression program in the developing brain

Gegenhuberet al 2022

What is the relationship between sex differences in the brain and sex-typical social behaviors



Source: DAALE 3

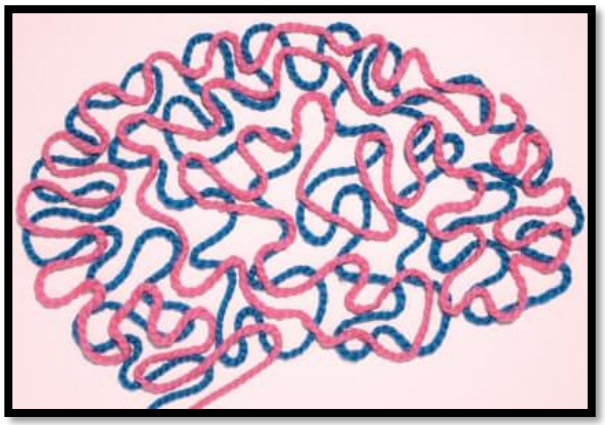
Male and female brains are weird differently?

Dimorphic brain (binary)



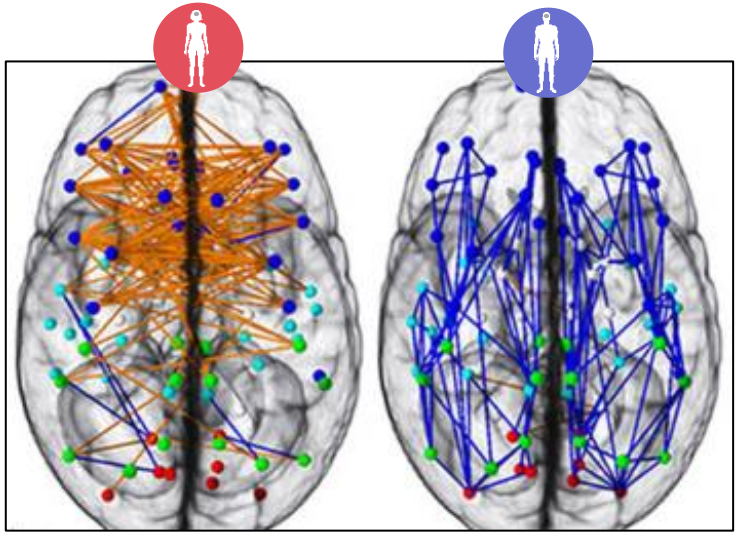
?

Mosaic brain



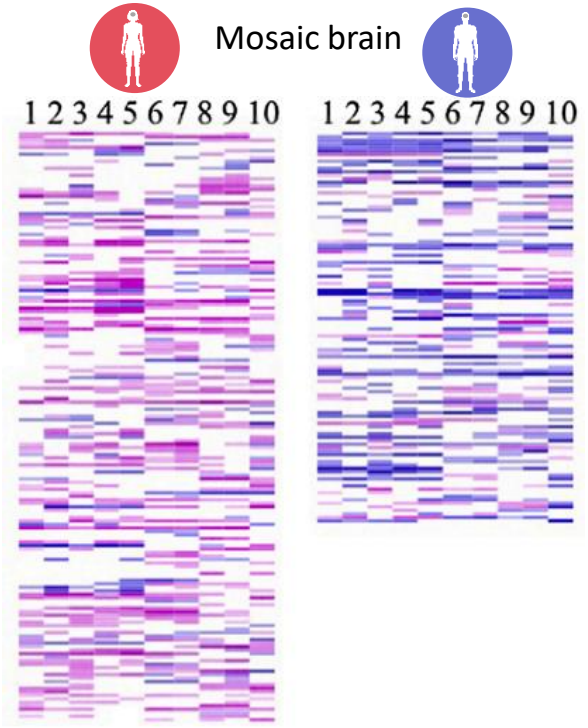
Dimorphic brain (binary) vs Mosaic brain

Dimorphic brain



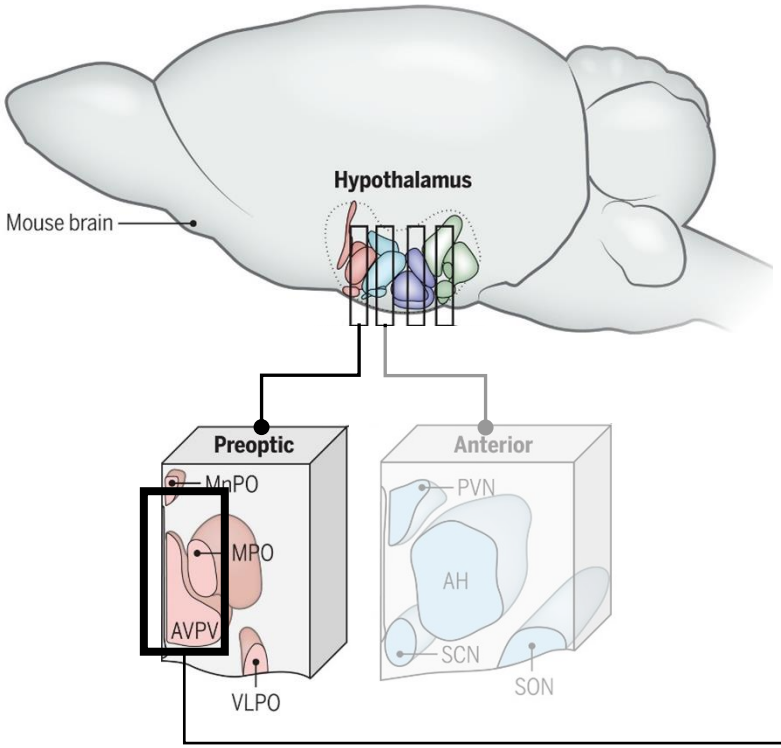
Ingalhalikar et al 2013, *PNAS*

Mosaic brain

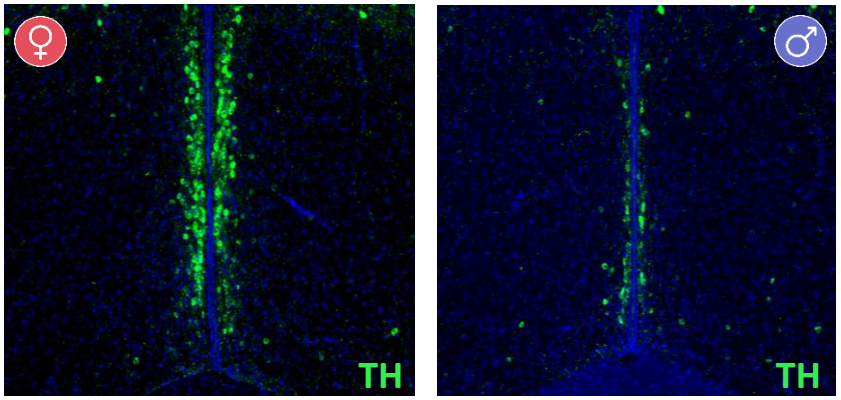


Shalev et al 2019, *Brain Struct & Funct*

Sex differences in gene expression in the Anteroventral Periventricular Nucleus (AVPV) of the hypothalamus



TH⁺ AVPV expressing neurons



TH= Tyrosine Hydroxylase (dopaminergic neurons)

Adapted from: Fong et al. Science, 2023

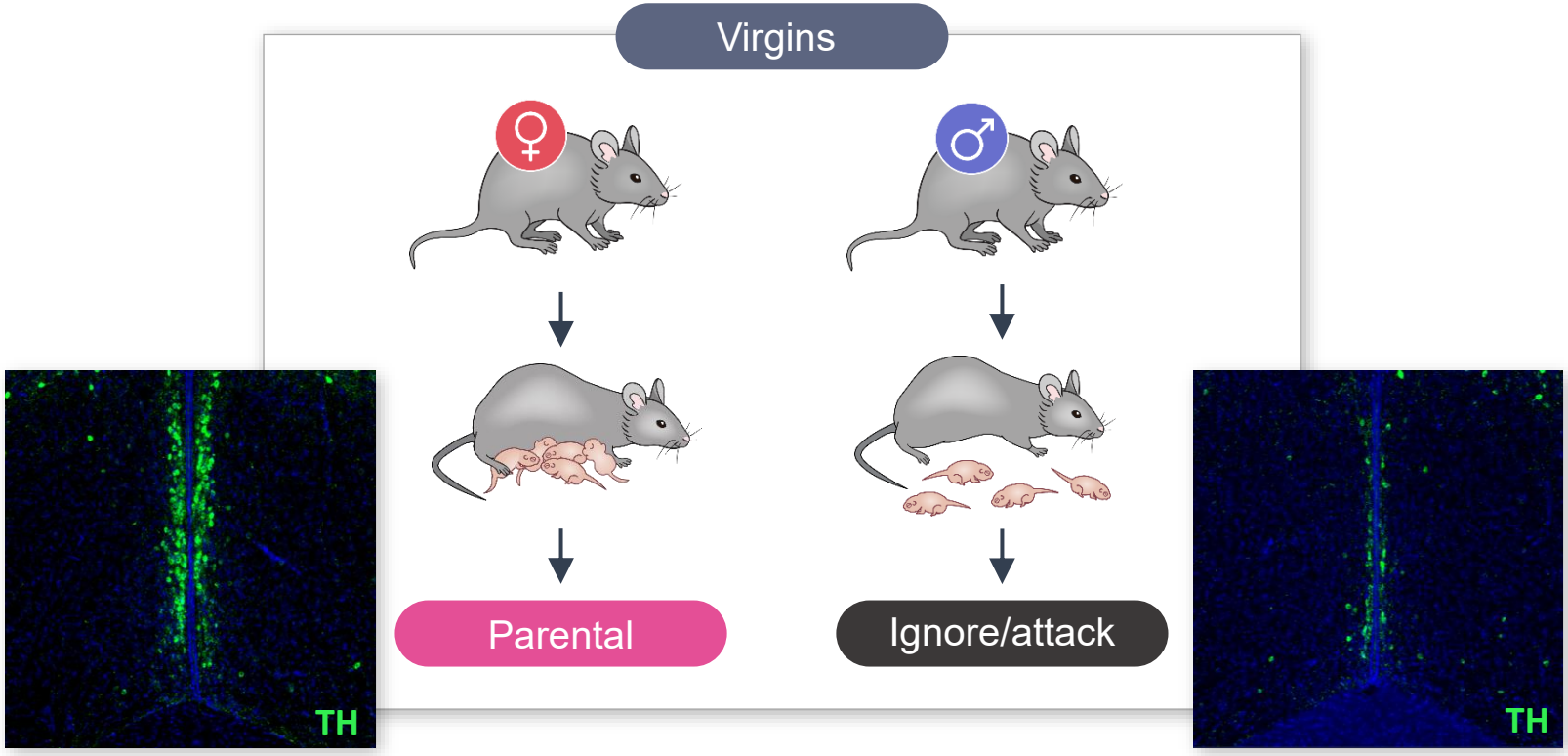
Parental care: Sexually dimorphic and evolutionary conserved








Sex differences in pup-directed behaviors in lab mice



Testing parental care in mice



Pup retrieval



Licking



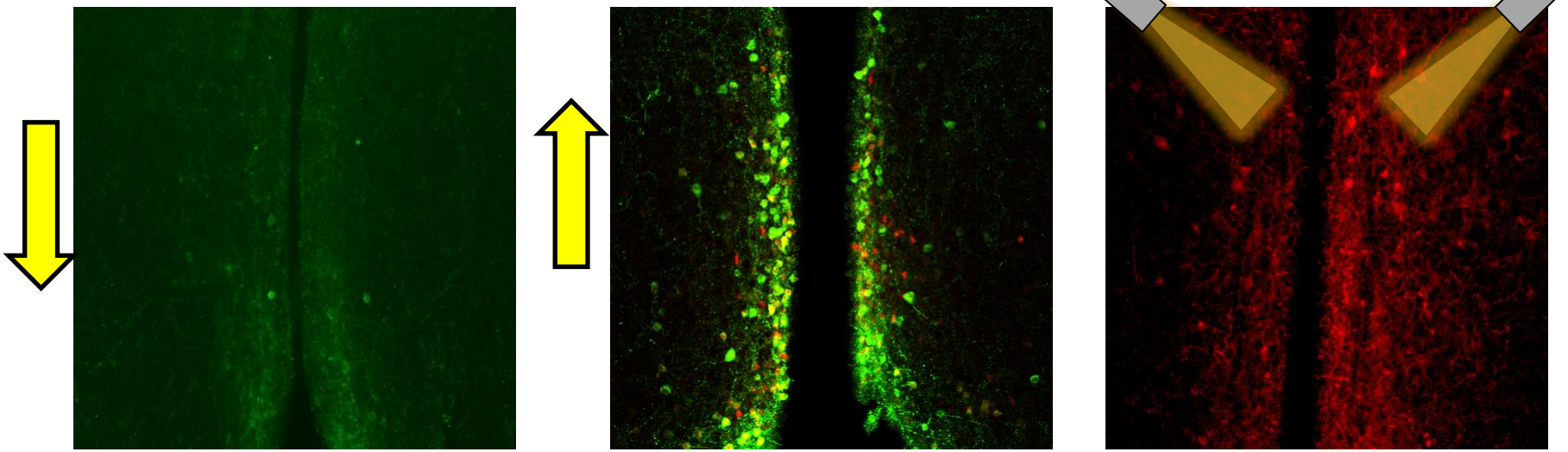
Nursing

Selective manipulations of TH^{+} AVPV neurons in adult males and females

Neuronal ablation

Neuronal over-expression

Neuronal activation using light stimulation (optogenetics)



TH-ablation

TH-overexpression

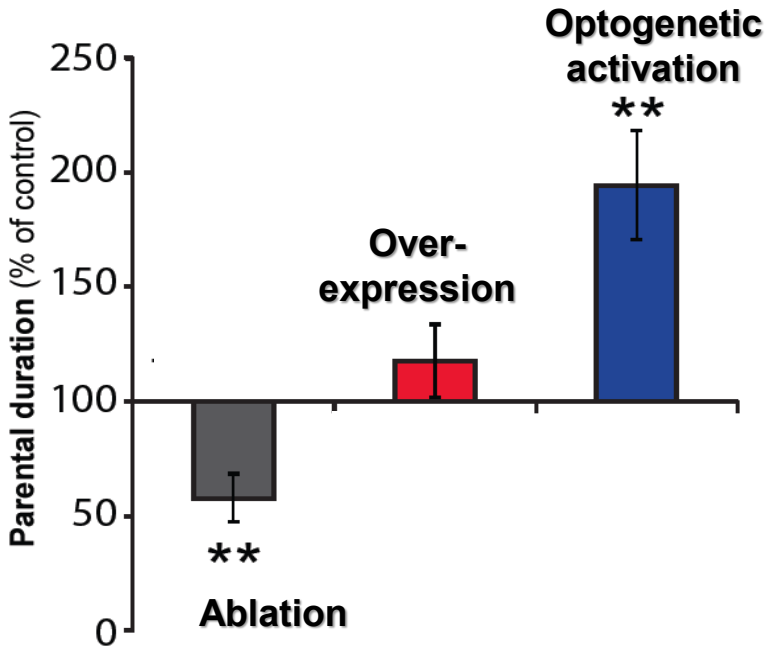
TH-ChR2

Scott et al. 2015

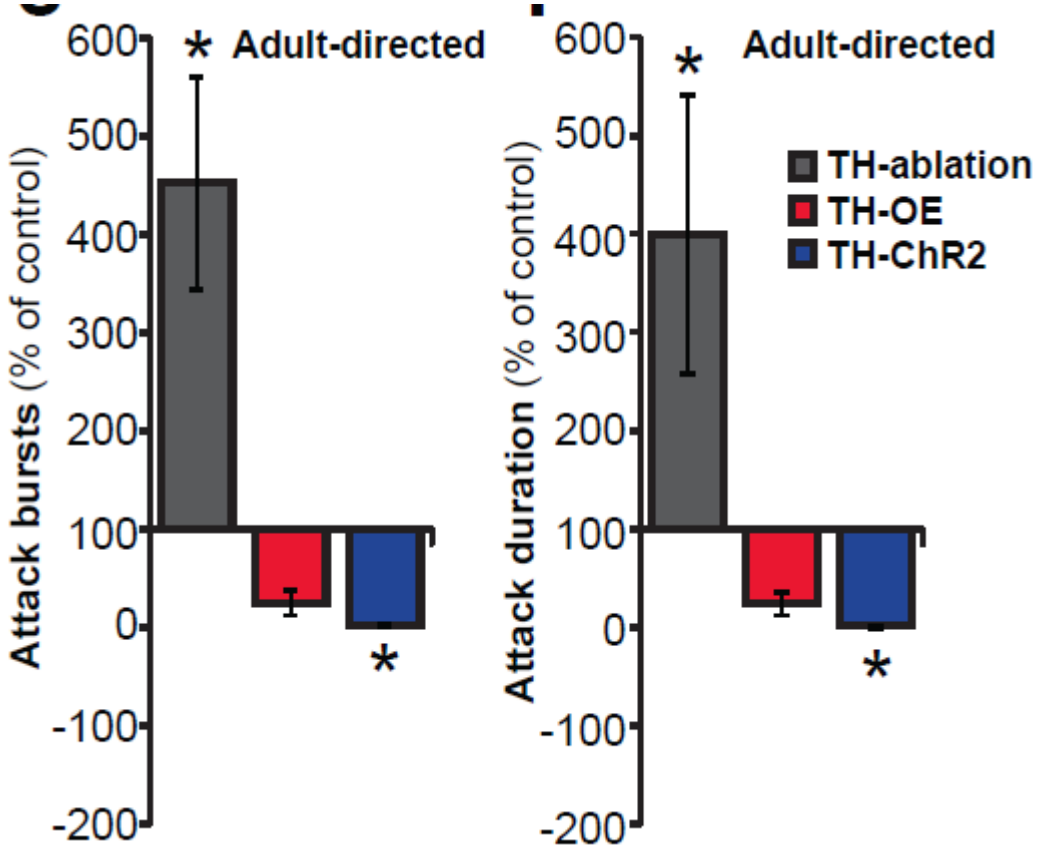
In females, hypothalamic dopaminergic (TH^+ AVPV) neurons promote maternal care





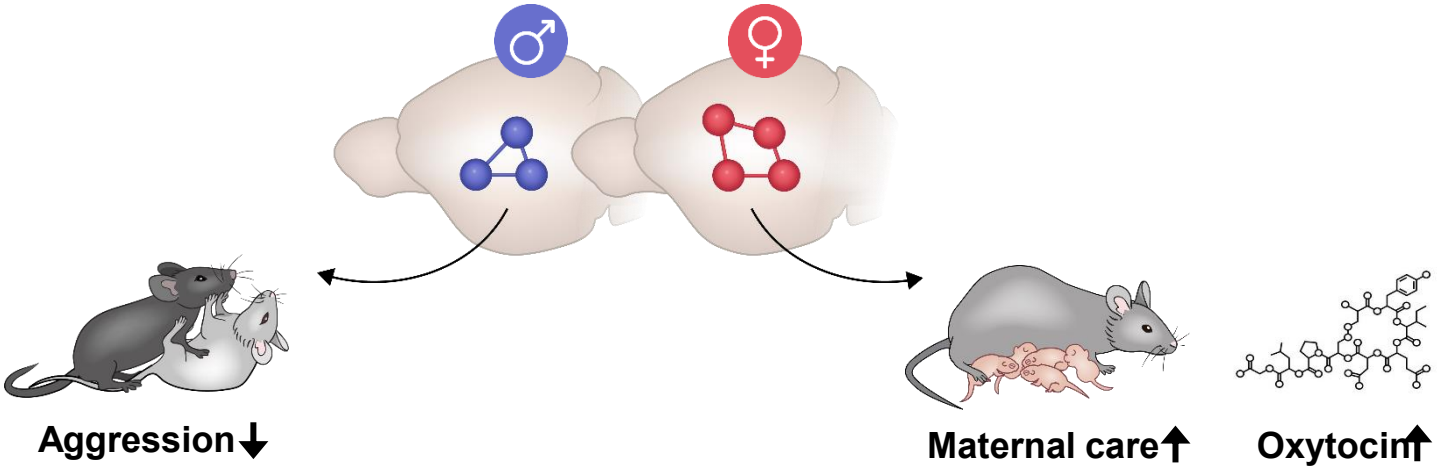


TH+ AVPV neurons are involved in suppression of conspecific aggressive behaviors

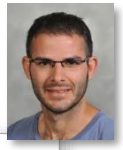


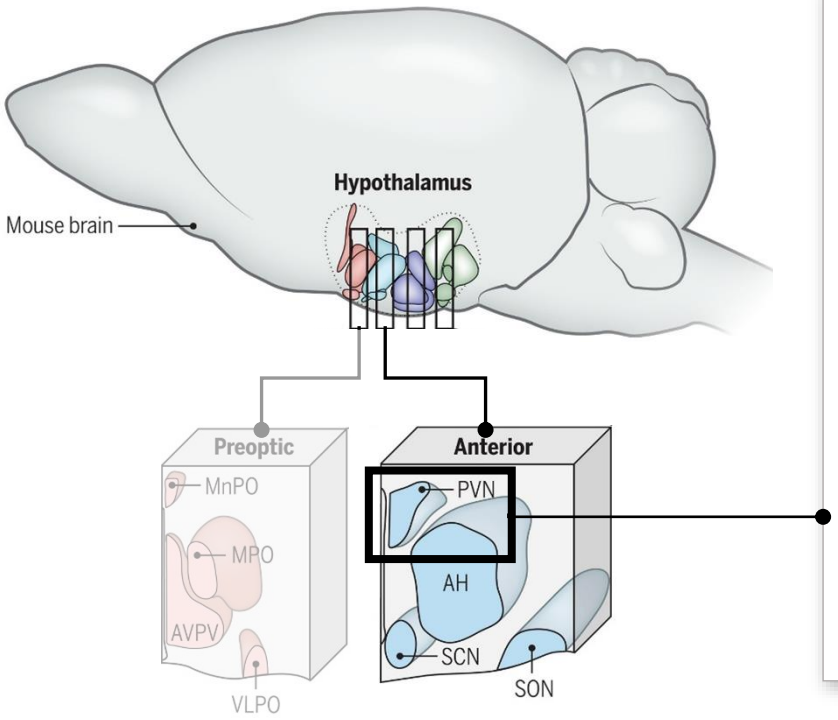
Sex-specific functions of sexual dimorphic dopaminergic neurons (TH^+ AVPV)



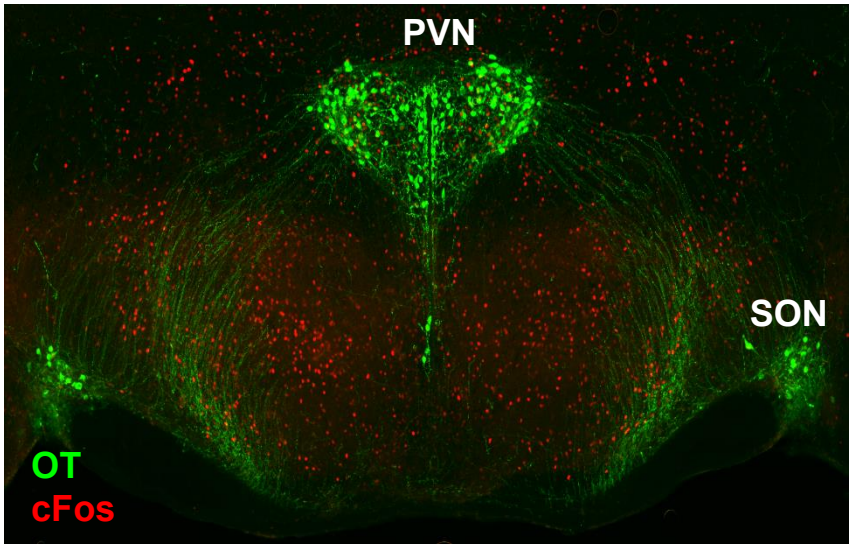


Studying the role of the oxytocinergic (OT) system in dominance hierarchy, in both males and females





Adapted from: Fong et al. Science, 2023



Sofer et al. Nat Neurosci, 2024

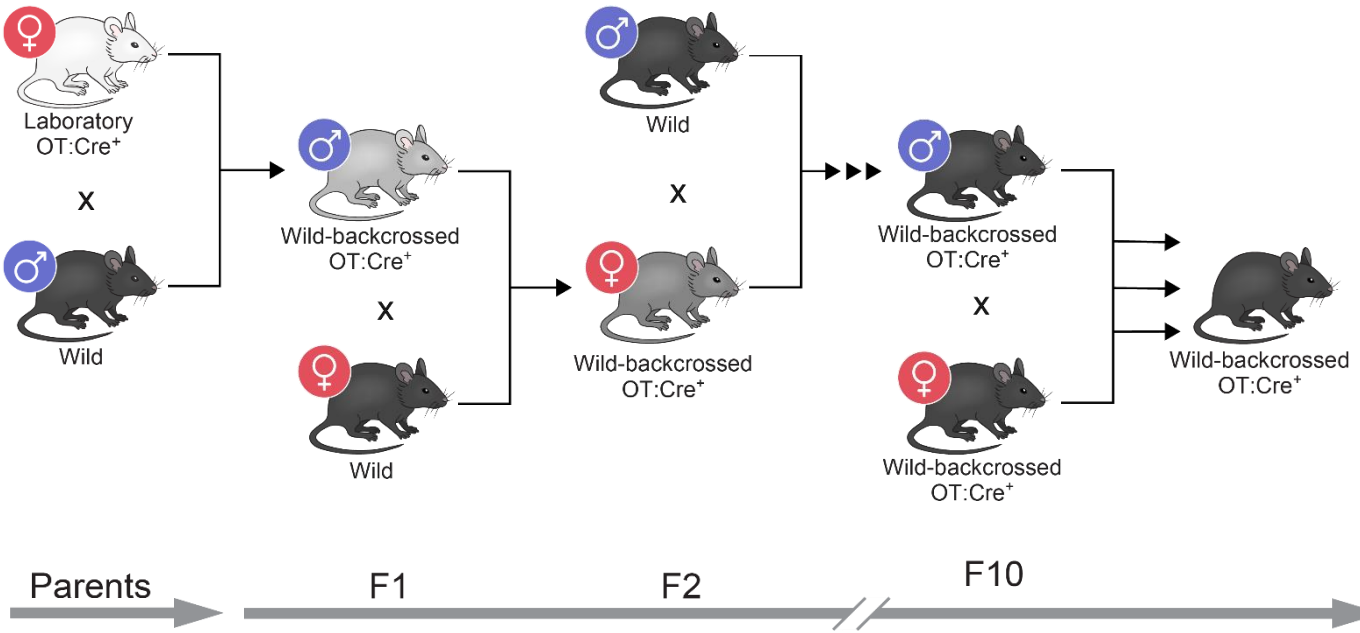
Tested individual: wild male and female house mice

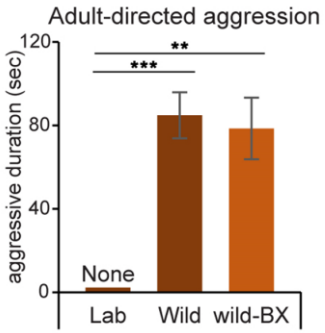
Robust aggressive behavior, in both males and females



Chalfin et al. Nat Comm, 2014
Zilkha and Kimchi, Curr Opin Neuro, 2016
Zilkha et al. Curr Biol, 2023

Generating wild-backcrossed OT:Cre mice to map and manipulate the oxytocinergic system

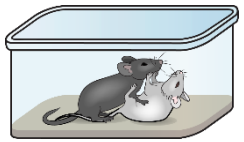




Sofer et al. Nat Neurosci, 2024

OT^{+} PVN neurons promote adult-directed dyadic aggressive behavior, only in females

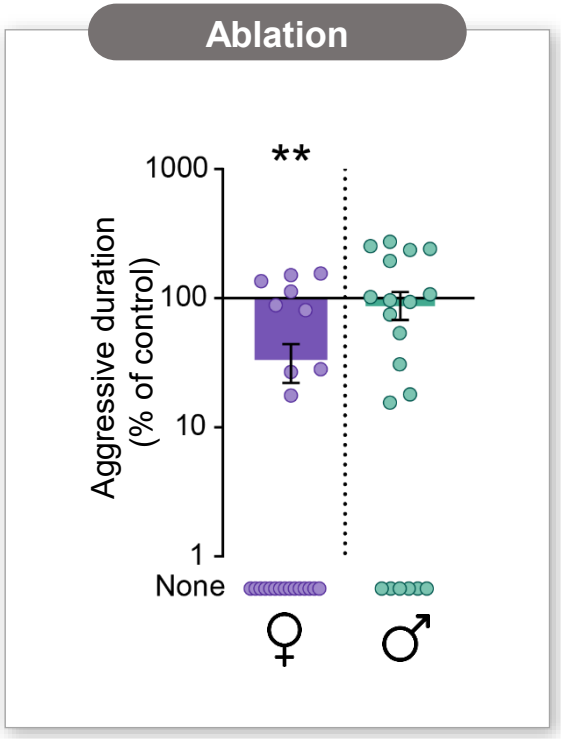
Wild-Backcrossed OT:Cre

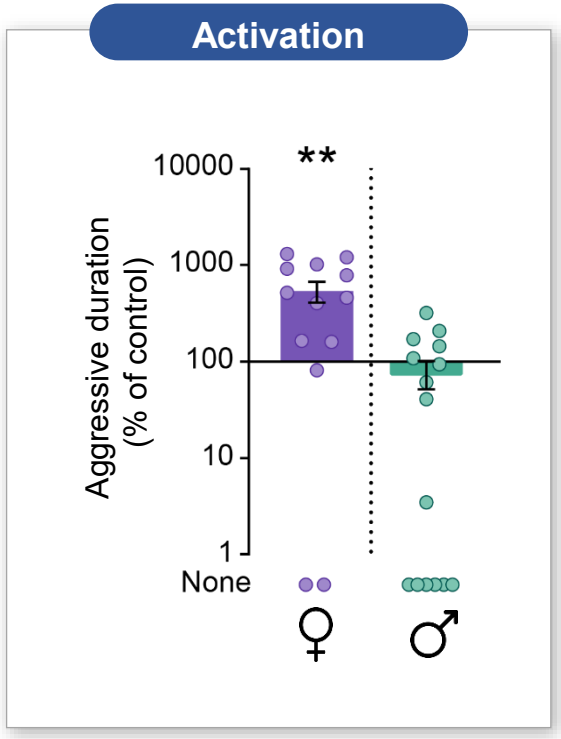


Control

OT-Ablation
AAV-DIO-taCasp3-Tevp

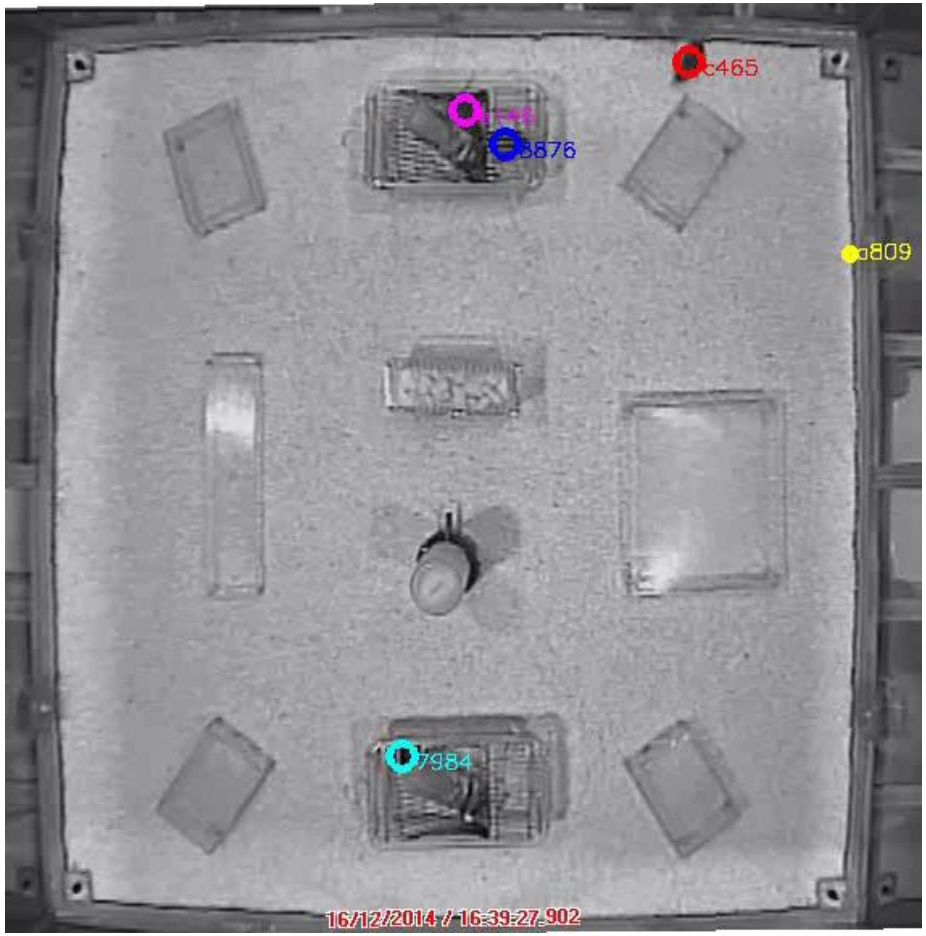
OT-Activation
AAV-DIO-hM3D-mCherry





Sofer et al. Nat Neurosci, 2024

Territorial aggression and competition in a group of wild females

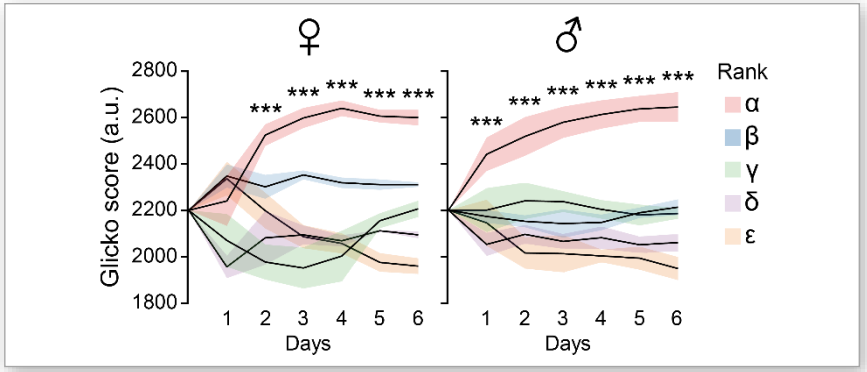




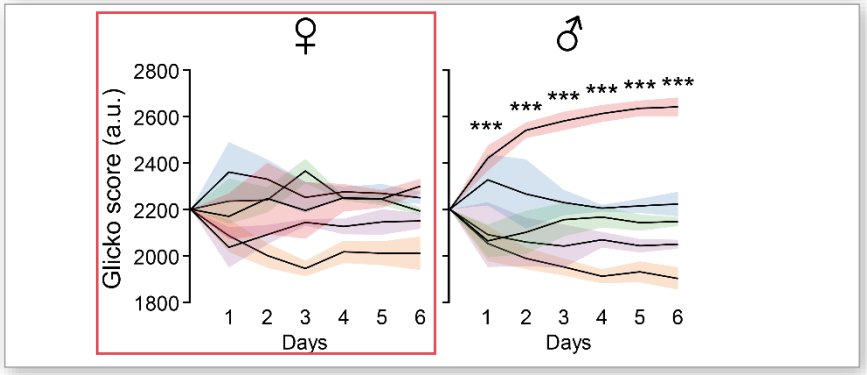
Ablation of OT⁺ PVN neurons abolishes dominance hierarchy, in females only





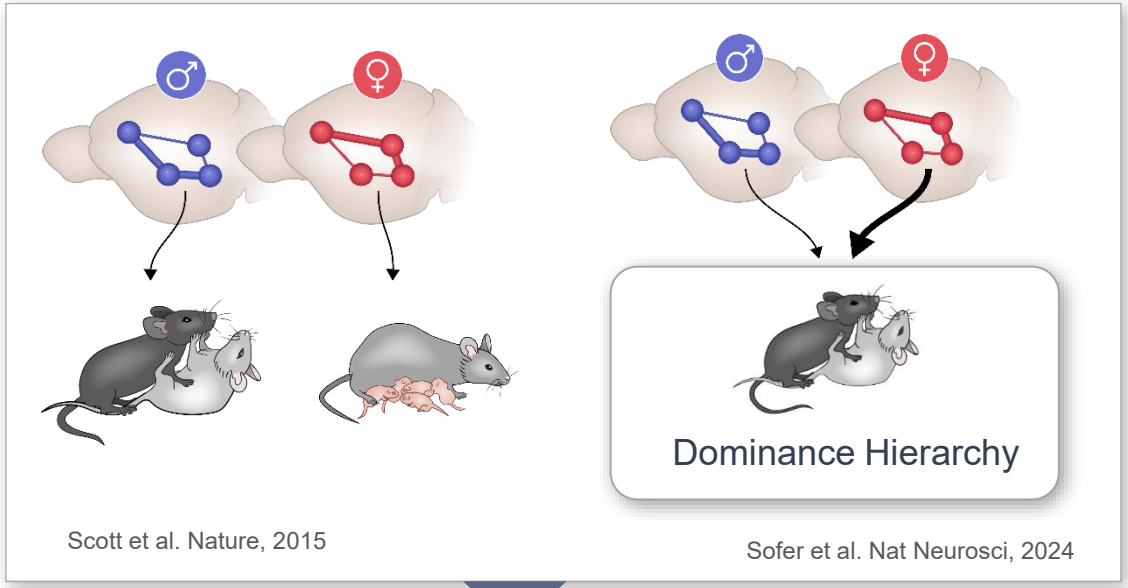






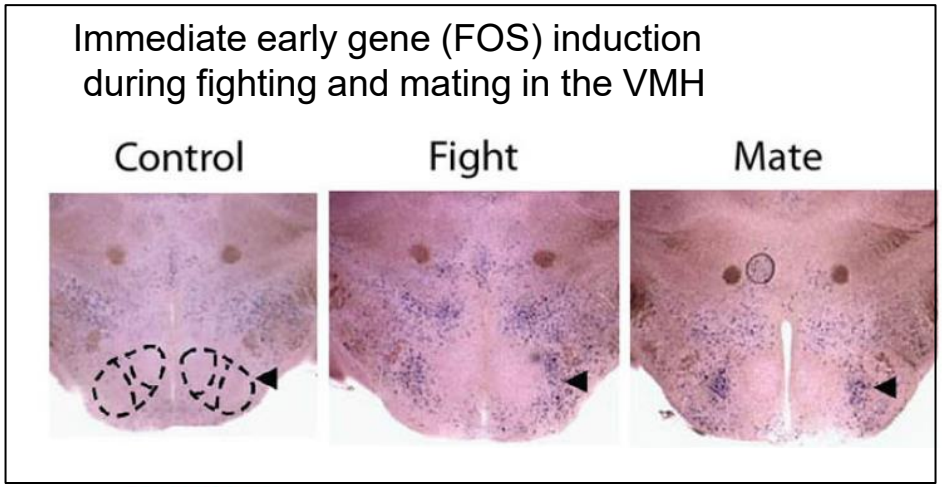
Sofer et al. Nat Neurosci, 2024

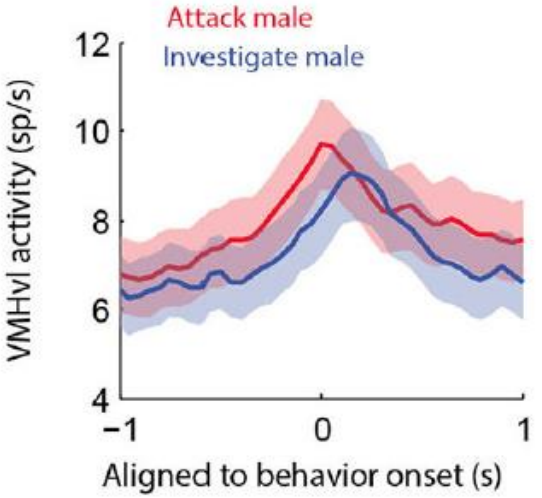
Sex differences in the brain do matter for social behavior

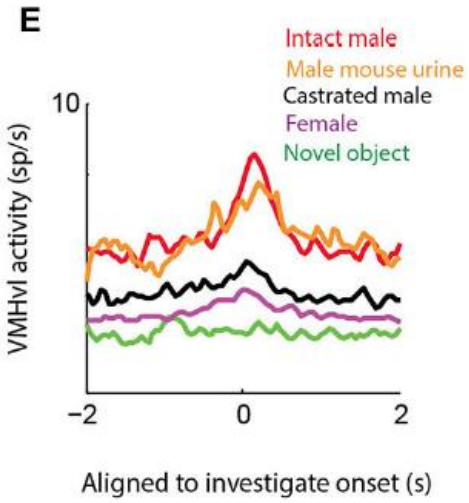


Sex must be considered
and compared

Functional identification of an aggression locus in the mouse hypothalamus (VMH)

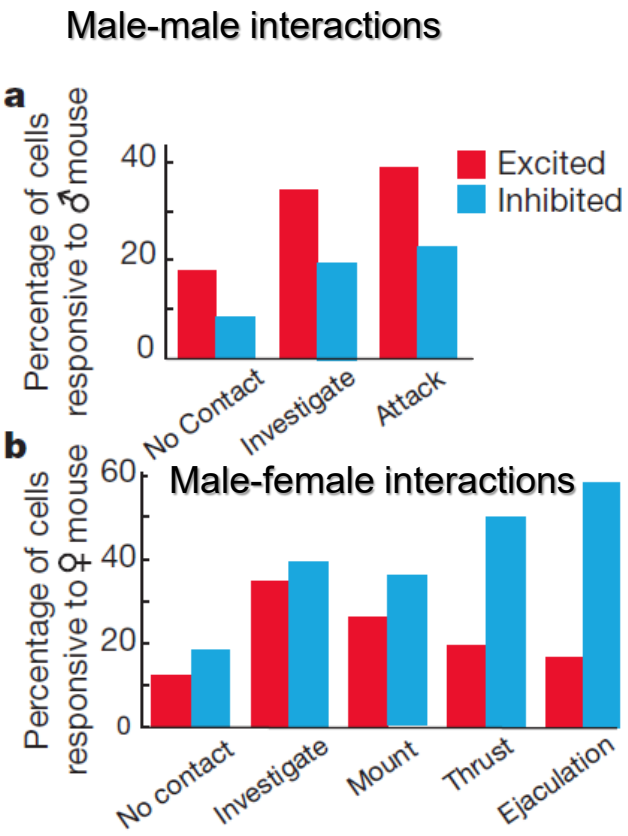


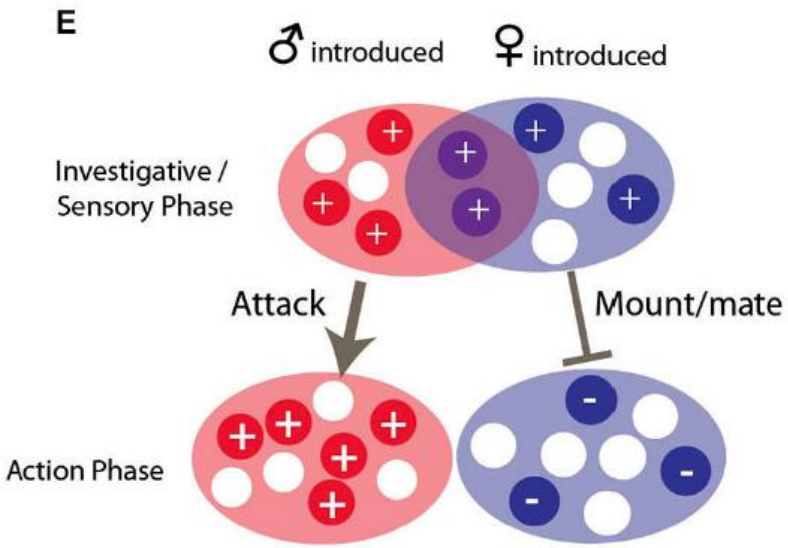




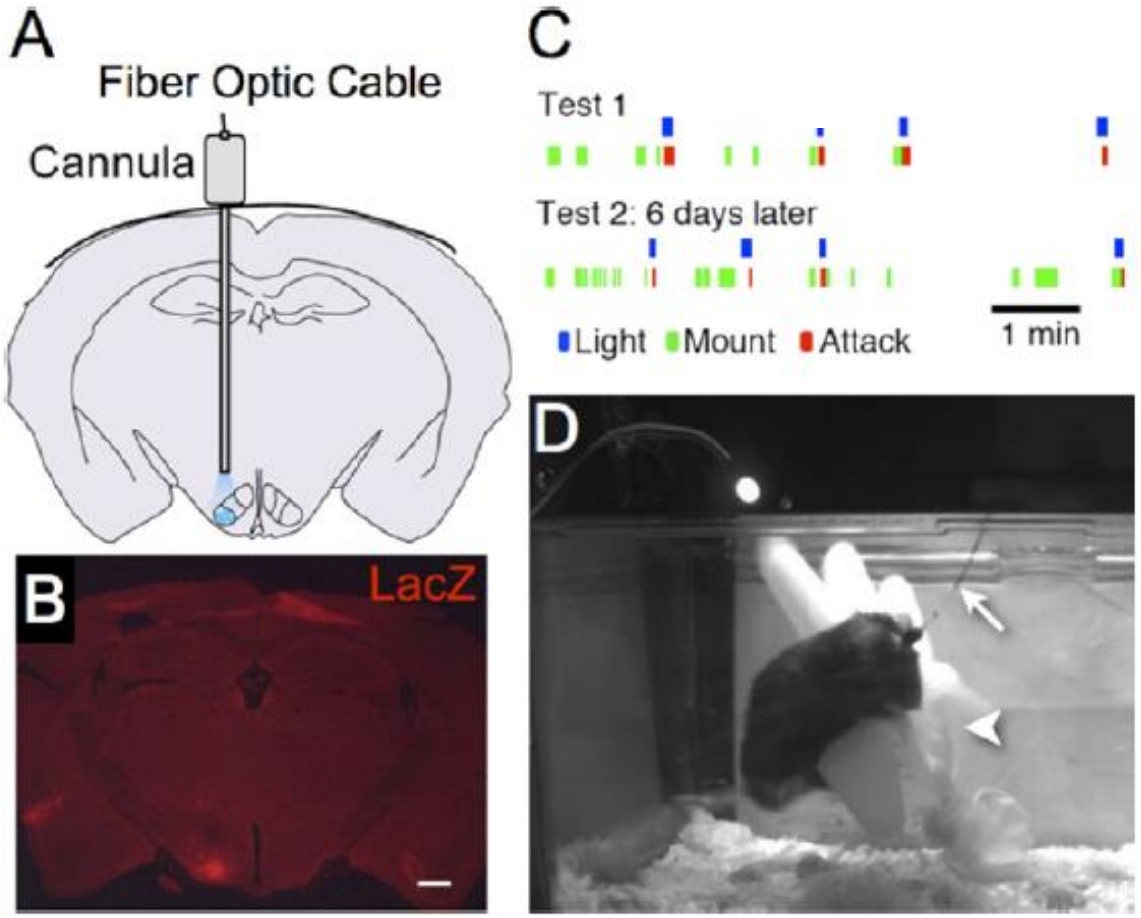
Lin et al 2011
Falkner and Lin 2014

Cell responses in VMH during mating and fighting





Activation of aggressive behavior using optogenetics in the VMH

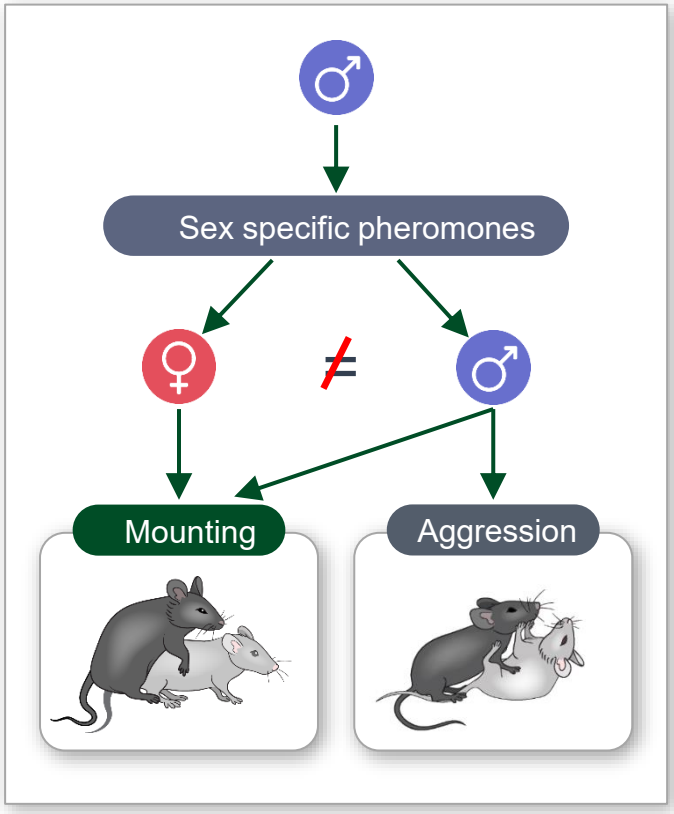


Lin et al 2011

The control of pheromone signals on sexually dimorphic reproductive behaviors

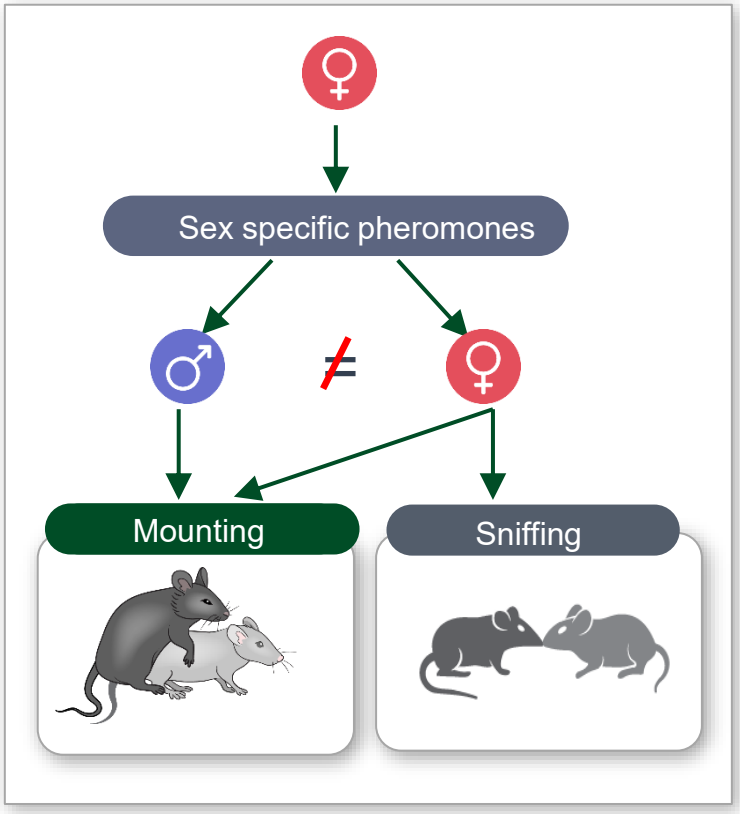


Mutation in a single gene (TrpC2) lead to loss of sexual discrimination in males



Stowers et al., Science, 2002

Mutation in a single gene (TrpC2) lead to loss of sexual discrimination in females



Kimchi et al., Nature, 2007

TRPC2$^{-/-}$ mutant females present male-typical sexual behavior



♀ **Mutant female**
(Light color)

♂ **Wild-type male**
(Dark color)

Kimchi et al., Nature, 2007

Sexual chasing of TRPC2$^{-/-}$ females of intruder males in semi-natural conditions



Mutant females
(Light color n=4)

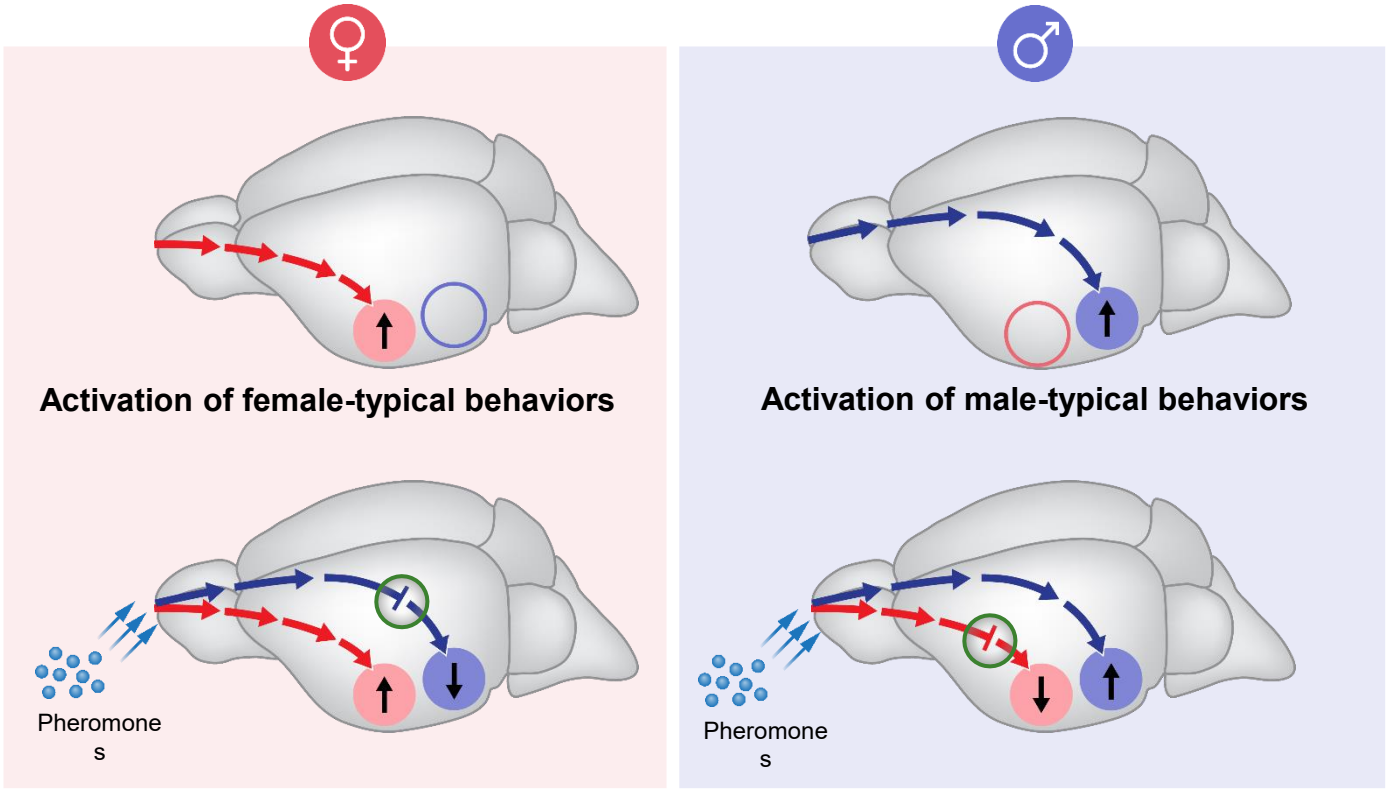
Wild-type males
(Dark color, n=2)

Kimchi et al., Nature, 2007

Sex-typical networks exist in males and females

Classical model

New model

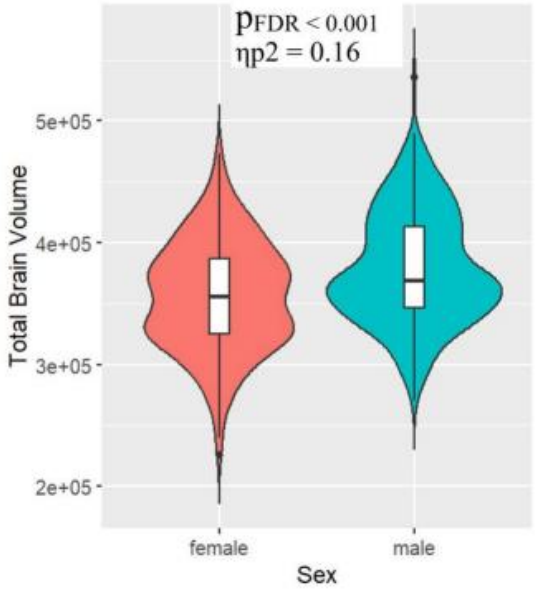


Kimchi et al., 2007, Nature

Human

Sex differences in the brain

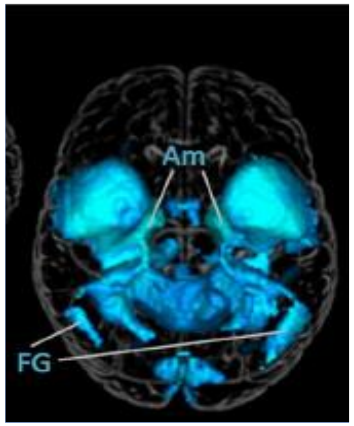
Total brain volume
M>F (~10%)



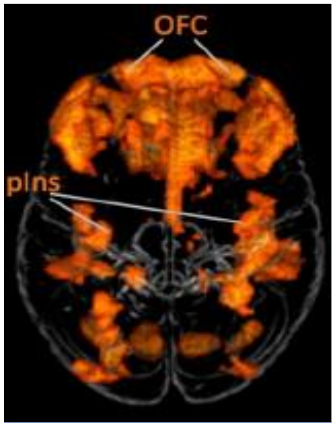
Kahn et al 2024, BMC

Differences in Gray Matter Volume
(M>F)

Men > Women

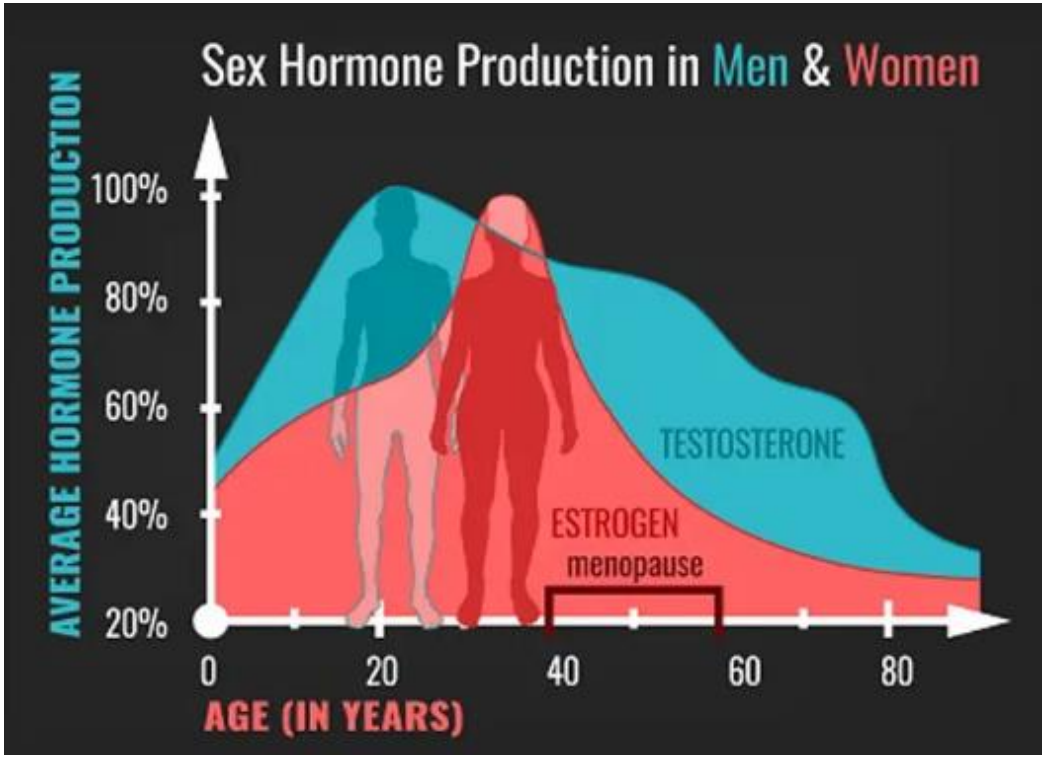


Women > Men

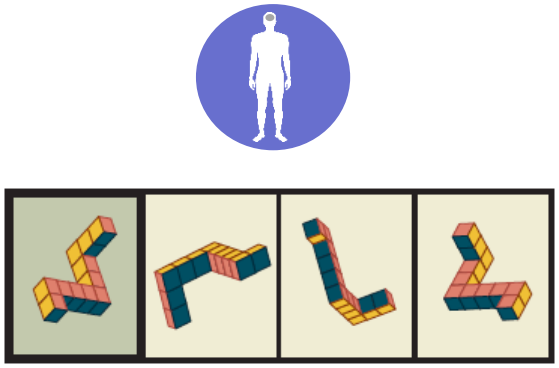


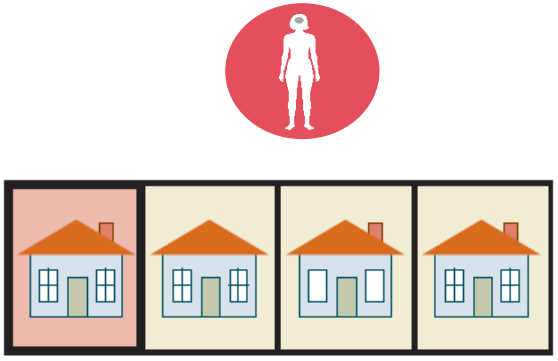
Lotz et al 2019, Sci Rep

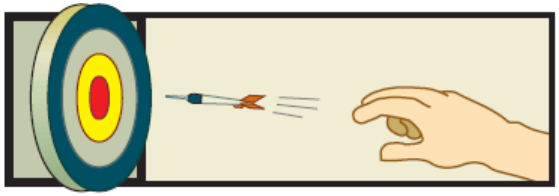
Sex differences in the hormones

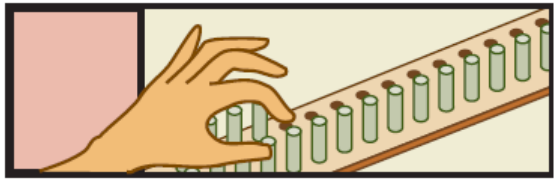


Sex differences in tasks



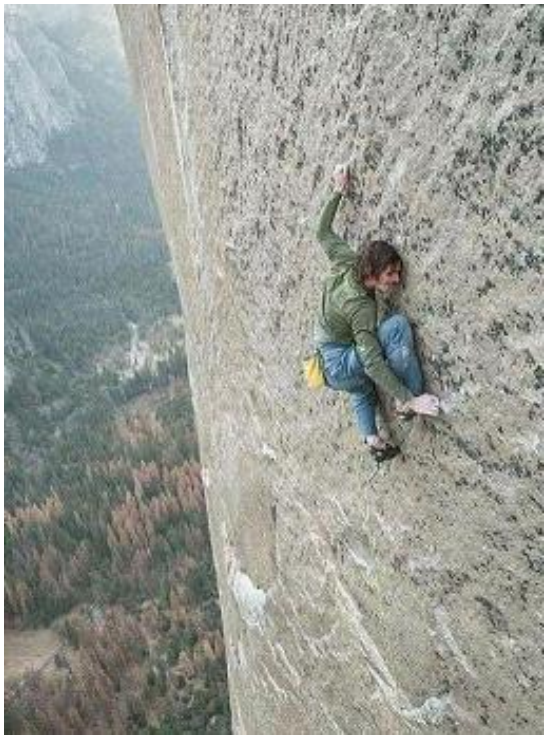




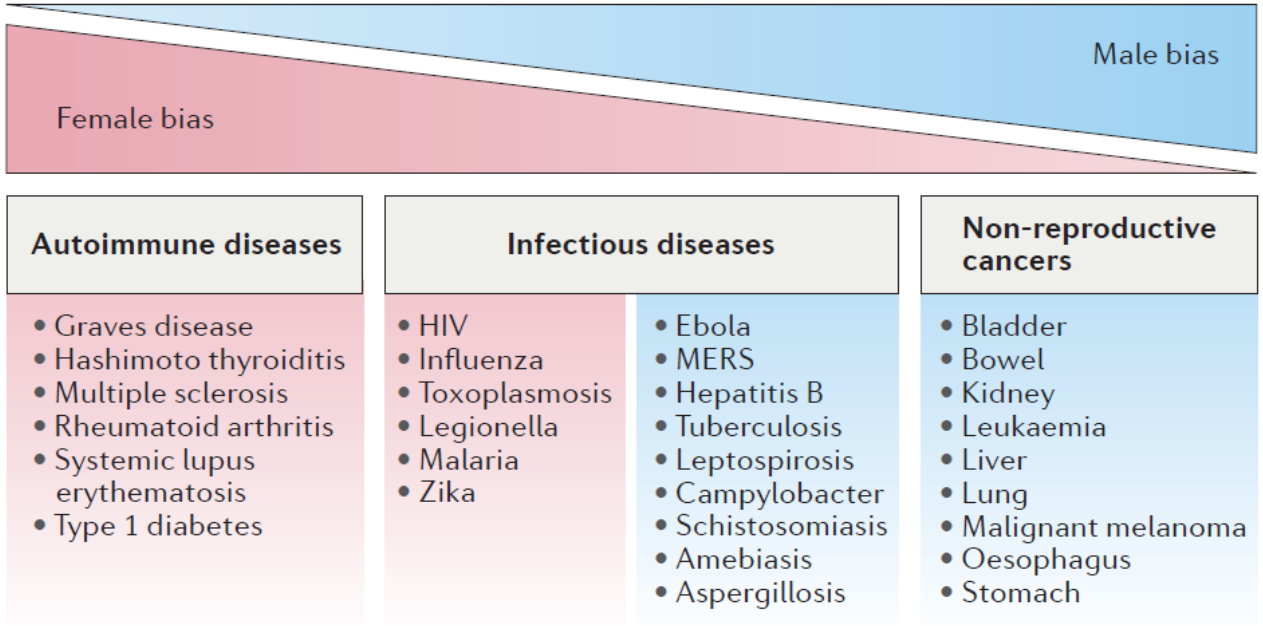


Risks taking; M>F





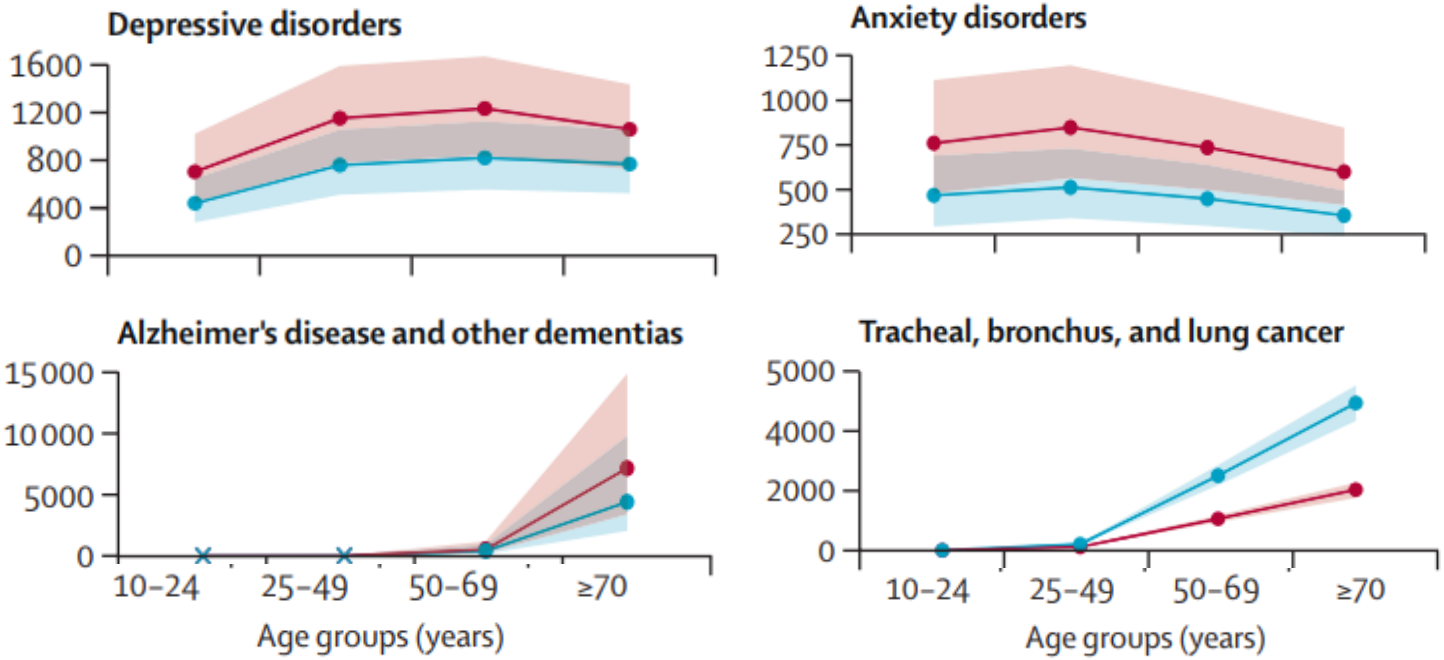
Sex differences in diseases



Klerin and Flangan, 2016, Nat Rev Immun

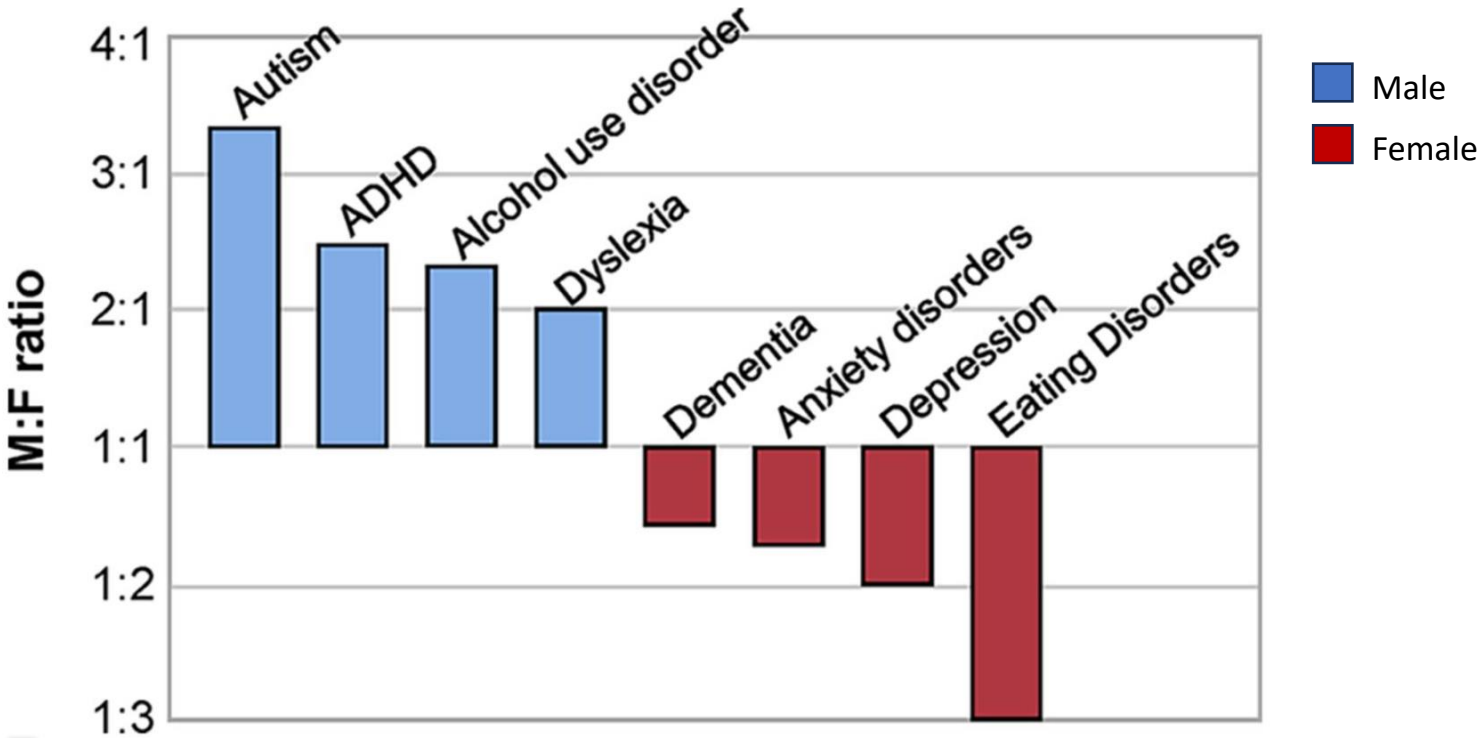
Women live longer but more sick





Patwardhan et al 2024, Lancet

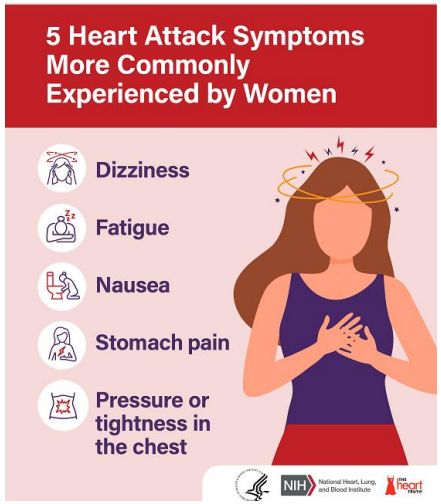
Sex differences in brain diseases



Brain Symptoms

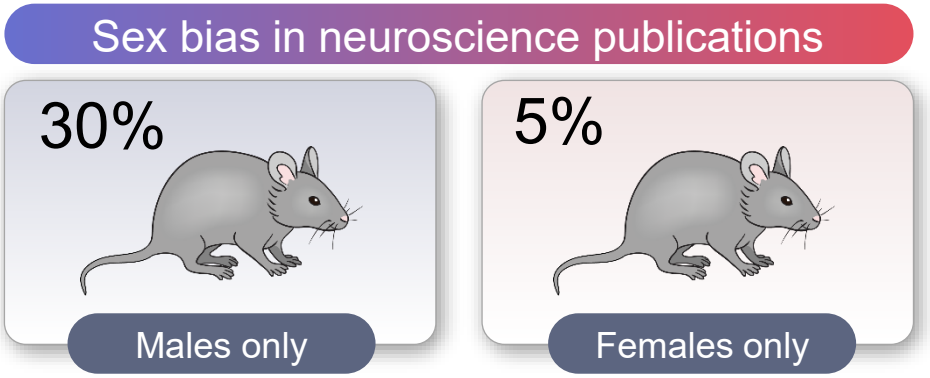


Women are more likely than men to die from heart disease (1:4 Women; 1:5 men)



Symptoms of heart attach is not same in men and women

Sex is often dismissed in neuroscience rodent studies



Zilkha and Kimchi, CONB, 2016
Will et al. eNeuro, 2017

Explanations why female data was not considered:

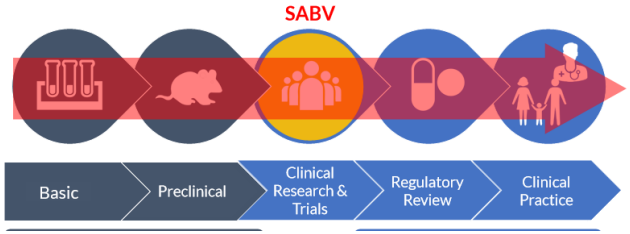
- Female data is **more variable** than data from males
- Testing females requires considering the **estrous cycle phase**
- We **did not find any sex differences**, so we report only male data

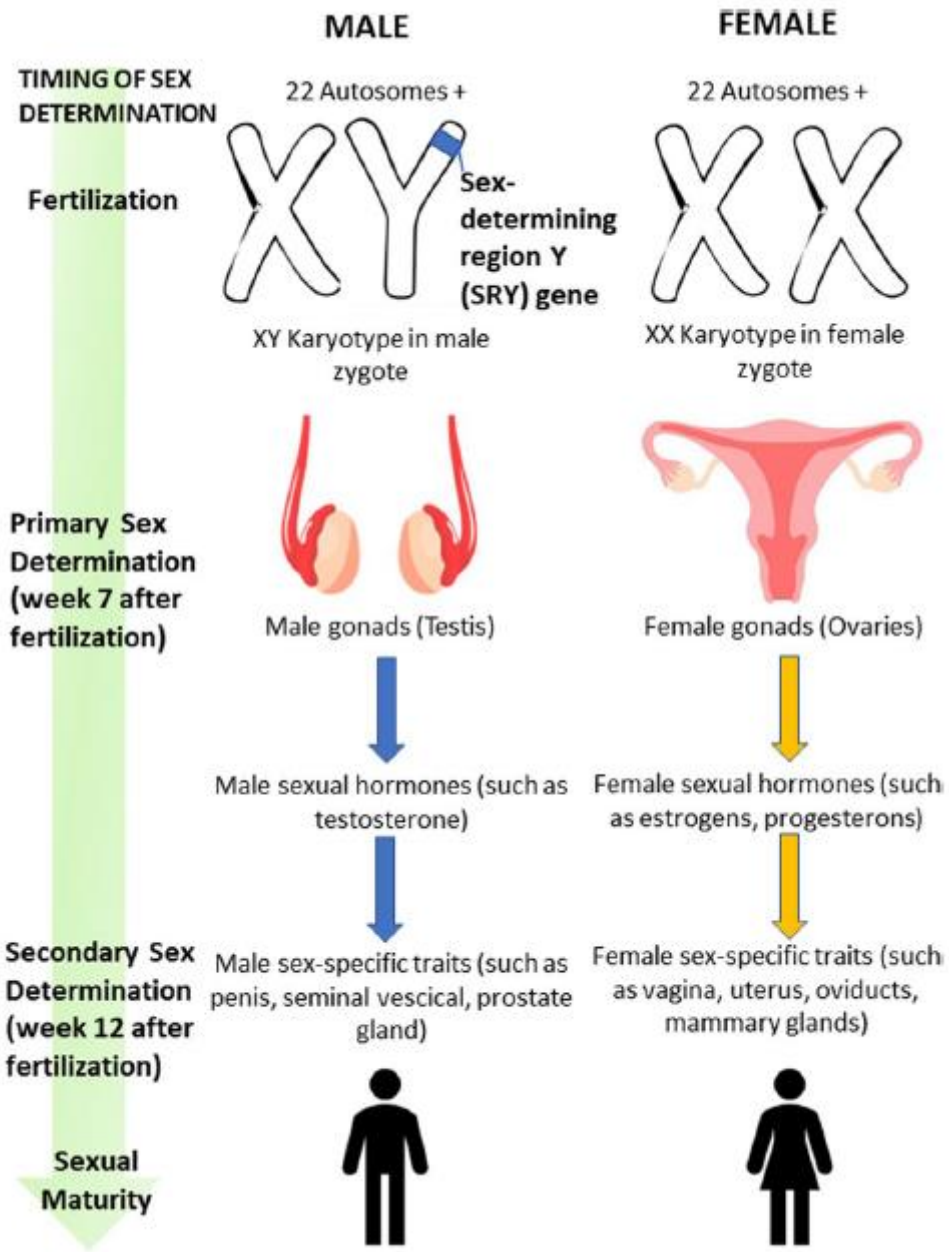
Shansky and Murphy, Nat Neurosci, 2021

Sex is a biological variable

- **1957-1962-** Thalidomide caused severe birth defects in over 10,000 children
- **1977-** FDA exclude women of reproductive potential from participating in early phase clinical research
- **1986 -** NIH recommends that women be included in medical research studies
- **2001** – FDA allows women of childbearing age to participate in early-phase clinical trials.
- **2016** –NIH mandates that sex be considered as a biological variable

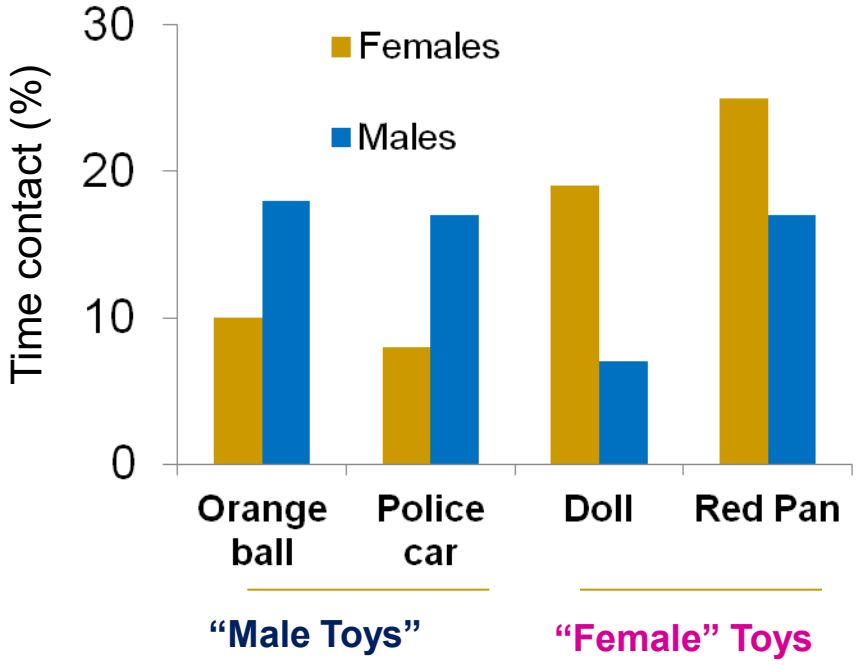
(SABV) in all animal and human studies.




MALE
FEMALE
TIMING OF SEX DETERMINATION
Fertilization
22 Autosomes +
22 Autosomes +
Sex-determining region Y (SRY) gene
XY Karyotype in male zygote
XX Karyotype in female zygote
Primary Sex Determination (week 7 after fertilization)
Male gonads (Testis)
Female gonads (Ovaries)
Male sexual hormones (such as testosterone)
Female sexual hormones (such as estrogens, progesterons)
Secondary Sex Determination (week 12 after fertilization)
Male sex-specific traits (such as penis, seminal vescical, prostate gland)
Female sex-specific traits (such as vagina, uterus, oviducts, mammary glands)
Sexual Maturity

Are there sex differences in behavior?





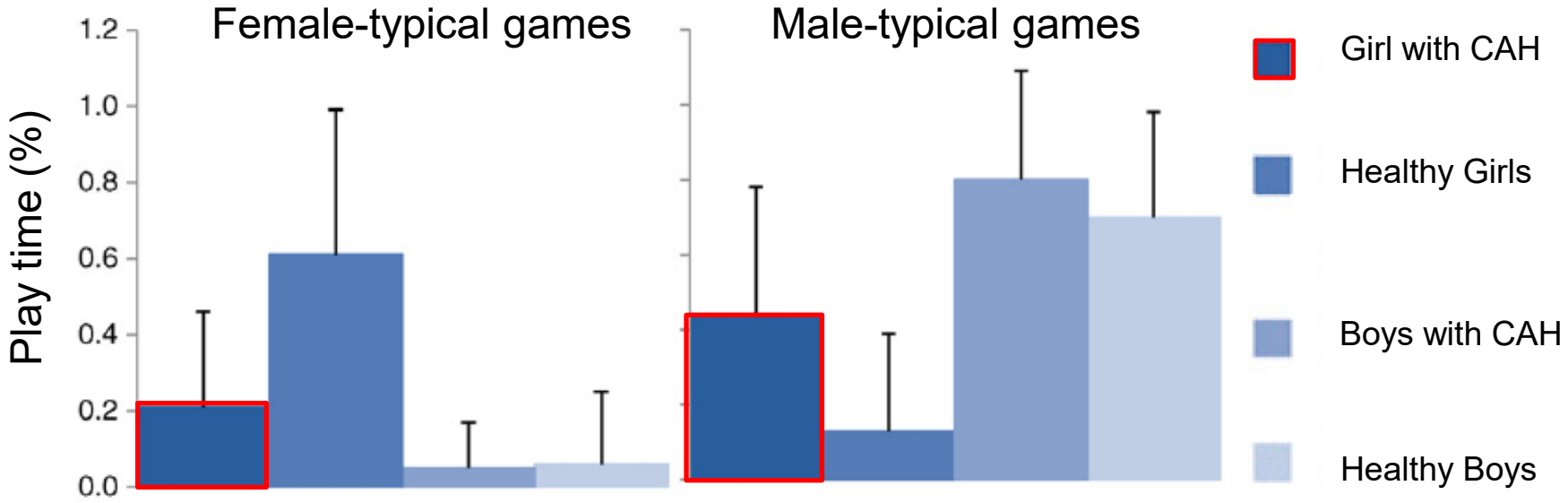


Alexander and Hines, 2002



Congenital Adrenal Hyperplasia (CAH)- Genetic disease

Elevated exposure to testosterone during development

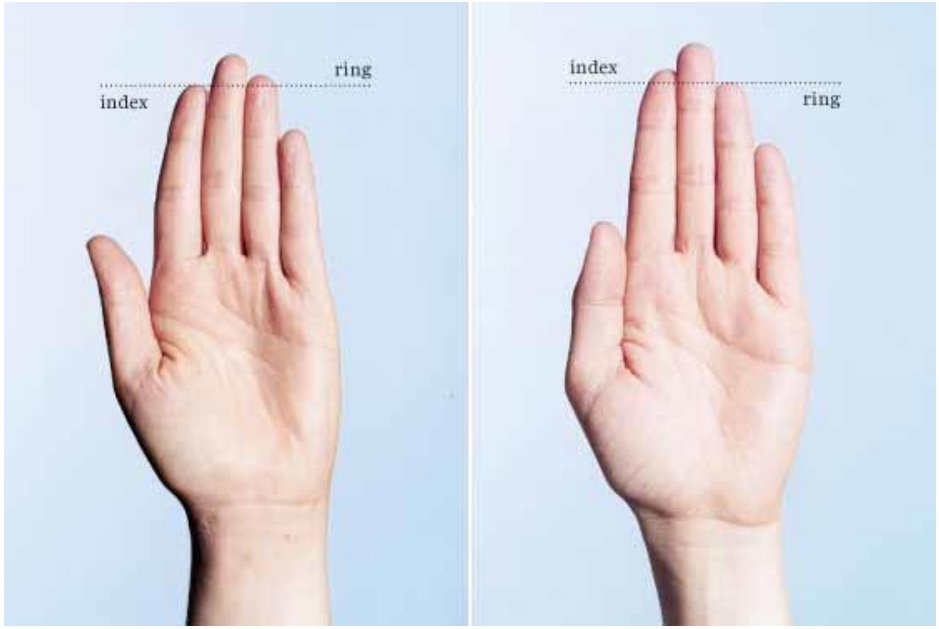






Pasterski et al 2005

Sex difference in digit ratio 2D:4D



Male-typical ratio Female-typical ratio



- The index fingers of most men are shorter than their ring fingers, and for most women they are the same length or longer.
- Right-hand 2D:4D might be a better indicator of prenatal androgenisation than left-hand 2D:4D.
- 2D:4D ratios may provide a useful retrospective marker for early androgen exposure, making it possible to correlate such exposure with human behavior.

Williams, T. J. et al (2000) Nature

Table 4 Correlations between left and right hand digit ratios and scores on the personality scales

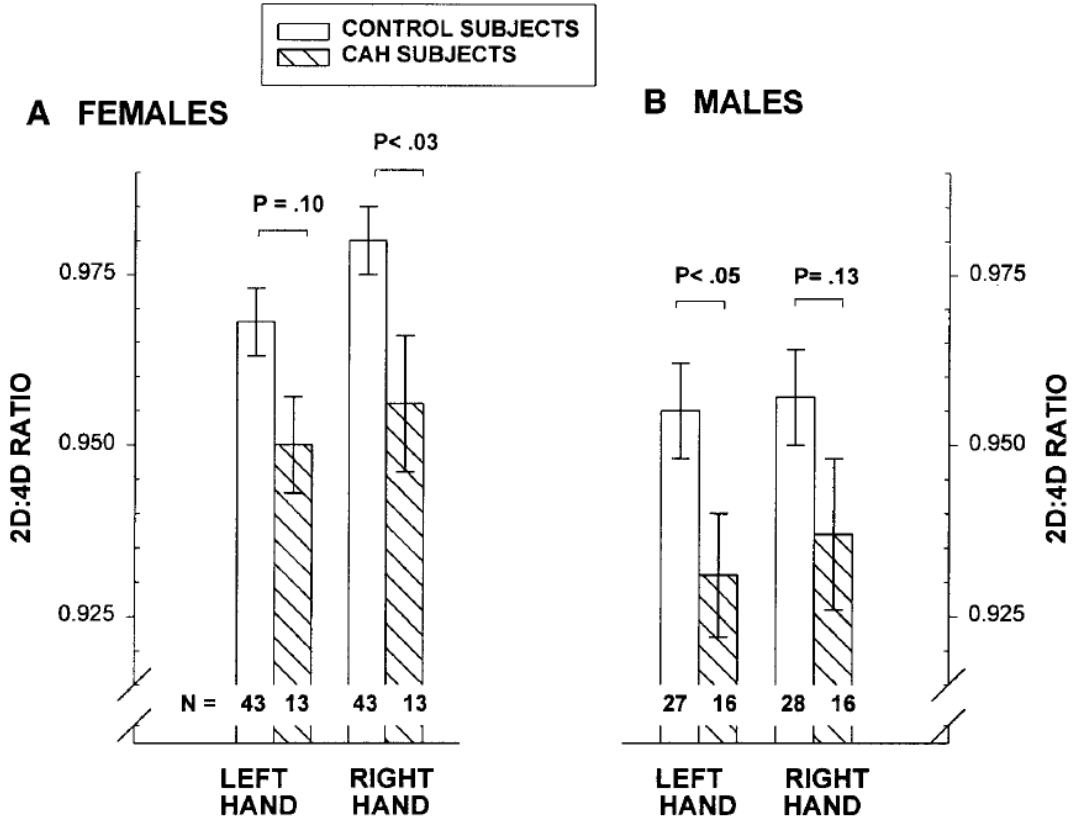
Scale	Men (n = 87)[a]		Women (n = 75)		Combined (n = 162)	
	Left r	Right r	Left r	Right r	Left r	Right r
Assertiveness	–.01	–.01	–.15	–.14	–.08	–.08
Emotional empathy	–.05	–.15	–.08	.11	.21***	.23***
PAQ masculinity	–.02	–.06	–.20*	.04	–.19**	–.11
PAQ femininity	–.15	–.05	–.09	–.21*	–.24***	–.22***
Nurturance	.14	–.01	–.17	–.12	.19**	.11
Sensation seeking	–.22**	–.10	–.10	–.23**	–.22***	–.22***
Boredom	–.19*	–.07	–.14	–.16	–.20***	–.14*
Disinhibition	–.12	–.12	–.04	–.13	–.12	–.14*
Experience seeking	–.13	.03	.05	–.14	–.05	–.05
Thrill/adventure	–.17	–.20*	–.16	–.25**	–.22***	–.25***
Aggression total	–.13	–.13	–.08	–.34***	–.15*	–.28****
Verbal	–.22**	–.22**	–.10	–.20*	–.20***	–.25***
Hostility	–.01	–.11	–.02	–.29**	–.03	–.20***
Physical	–.04	–.01	–.06	–.31***	–.14*	–.23***
Anger	–.17	–.10	–.11	–.29**	–.11	–.18**

****$p < .001$, ***$p < .01$, **$p < .05$, *$p < .10$

[a] Ns reduced on some scales due to missed items

Hampson et al 2008, Arch Sex Behav

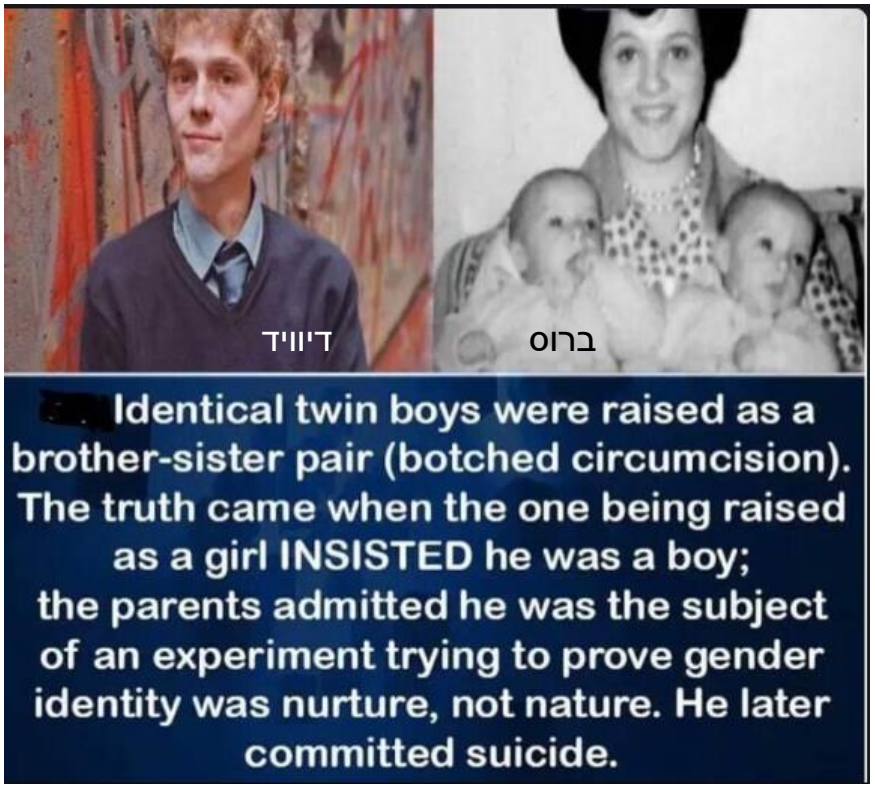
Masculinized Finger Length Patterns in Human Males and Females with Congenital Adrenal Hyperplasia



Brown et al 2002, Horm and Behav

The Boy Who Was Turned Into a Girl





https://youtu.be/dXFjmjC7Uos